GEK-65425

DRIVE SYSTEM INSTRUCTIONS

SPEEDTRONIC*
INTEGRATED TEMPERATURE SYSTEM SELF-VERIFICATION MODULE

PART NUMBER 277A5732G5

* TRADEMARK OF GENERAL ELECTRIC CO., U.S.A.

DRIVE SYSTEMS DEPARTMENT
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GENERAL ELECTRIC

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SECTION 1
GENERAL DESCRIPTION

1-1 SCOPE

This instruction manual describes the Self Verification Module (SVM) and explains its operation.

1-2 PURPOSE AND USE

The Self Verification Module (DS3820SVM) provides a means of testing and troubleshooting the hardware contained in a SPEEDTRONIC* Integrated Temperature System. The module contains hardware diagnostic programs in ROM (read-only memory) designed to assist the operator in verifying the correct operation of the Integrated Temperature System (ITS) hardware and in identifying defective printed wiring boards.

(a) Application

This publication is applicable to the program supplied in the SVM and ordered as a part of the equipment described herein (G.E. Part Number 277A5732G5). The part numbers of ROM's supplied with this module are 68A9172P0073 and 68A9172P0074.

(b) Intended User

The SVM should be used by personnel trained in its operation as well as in the SPEEDTRONIC* control system and associated documentation.

1-3 GENERAL DESCRIPTION

The Self Verification Module (SVM) is housed in a small box-type enclosure containing the module's components. See Figure 1-1. A black lock-in lever, located at the bottom of the front panel, is linked to the test receptacle which mechanically and electrically connects the SVM to the equipment under test.

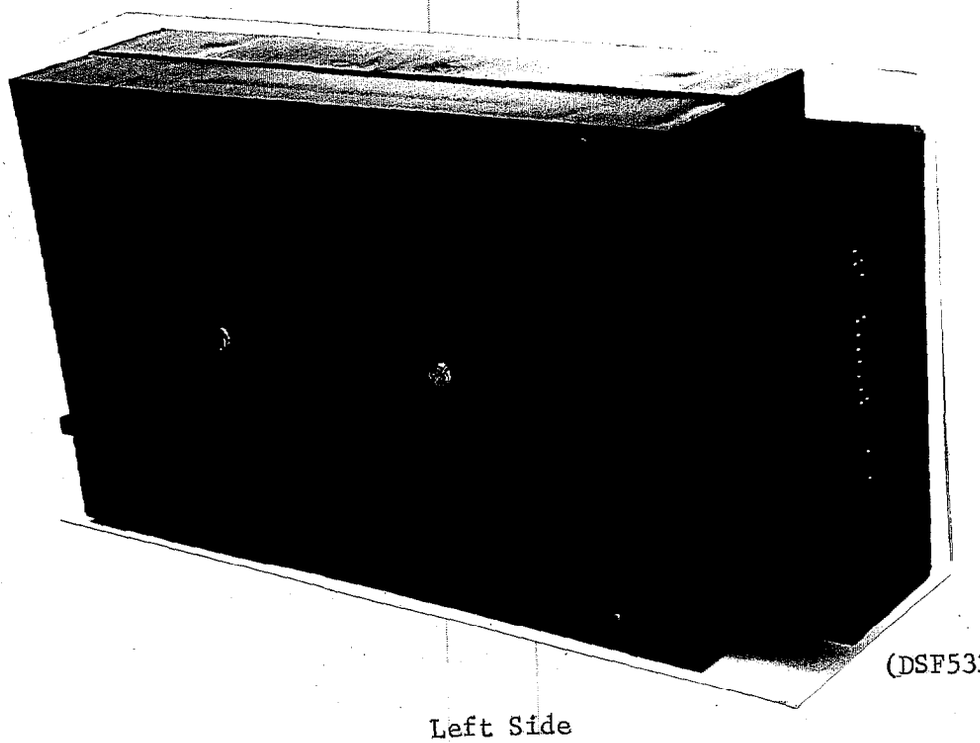
The SVM is supplied with a plastic bag for storage.

1-4 REFERENCE DATA

Size:	Length	8 inches (12.3 cm.)
	Height	3 7/8 inches (9.9 cm.)
	Width	1 3/4 inches (4.4 cm.)

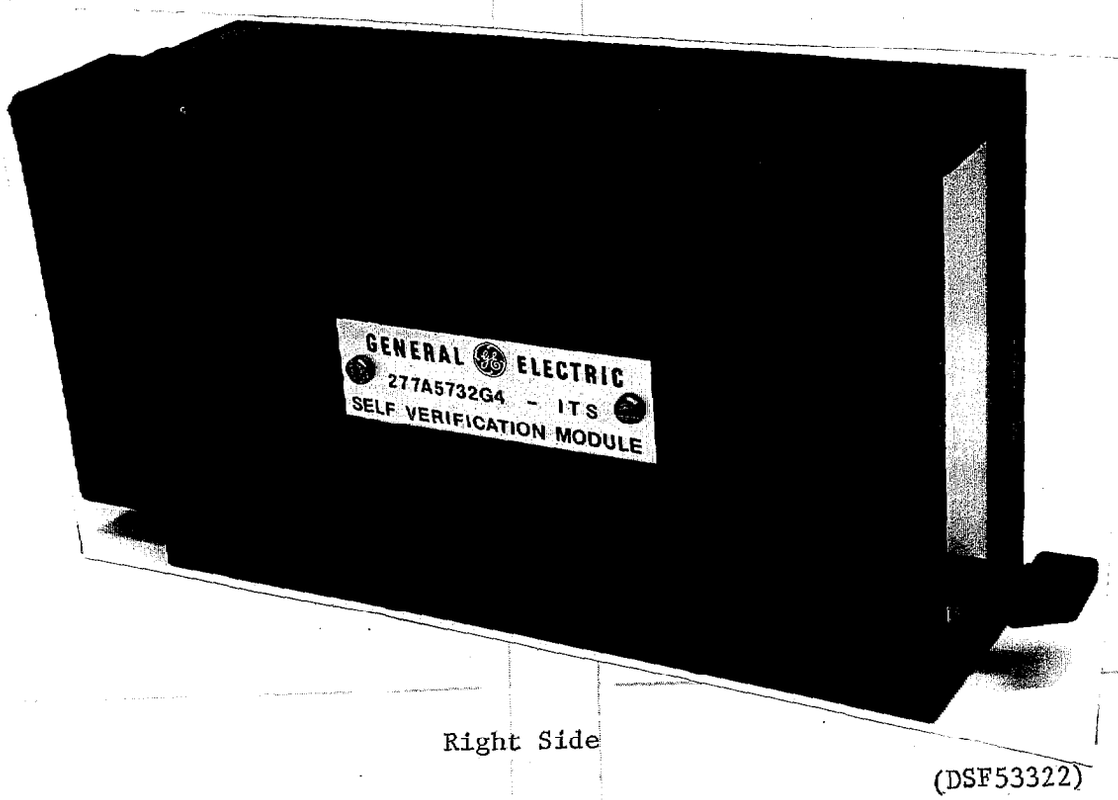
Weight: 1.3 pounds (0.6 kg).

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(DSF53323)

Left Side



Right Side

(DSF53322)

Figure 1-1 Typical Self Verification Module

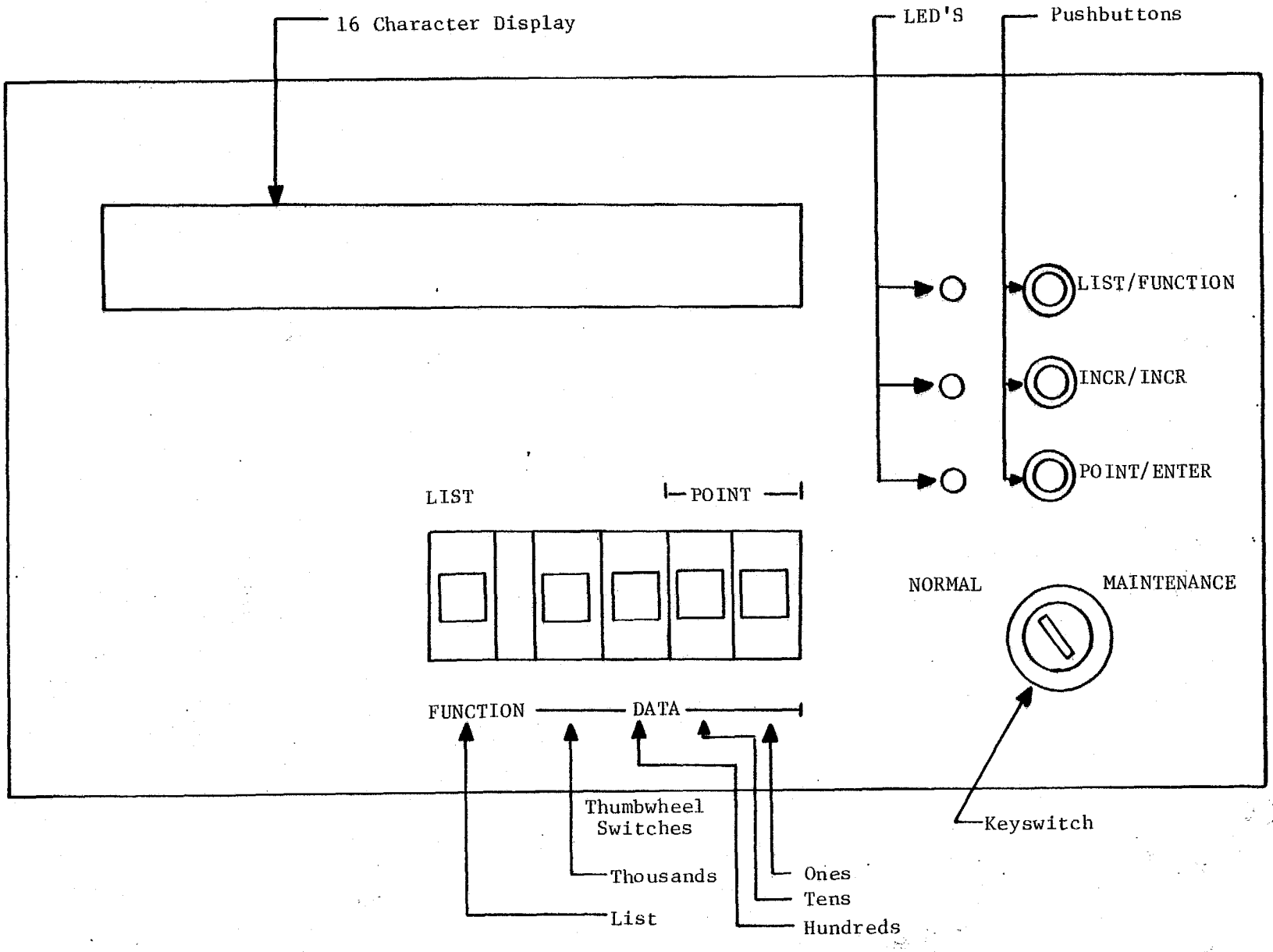


Figure 1-2. DS3820CDA Display Module Front Panel Layout

1-5 RELATED PUBLICATIONS

There are no publications directly related to the operation or use of the SVM.

1-6 DESCRIPTION OF CONTROLS, INDICATORS AND TEST POINTS.

There are no controls, indicators or test points located on the SVM. All controls and indicators used in conjunction with the SVM for ITS test purposes are located on the DS3820CDA Display Module (See Figure 1-2) and on the DS3800HMPA microprocessor card (HMPA). The display module contains controls, indicators and a 16-character display. The HMPA card contains light emmitting diodes (LED's) used for test purposes. Figure 1-3 shows a system test block diagram.

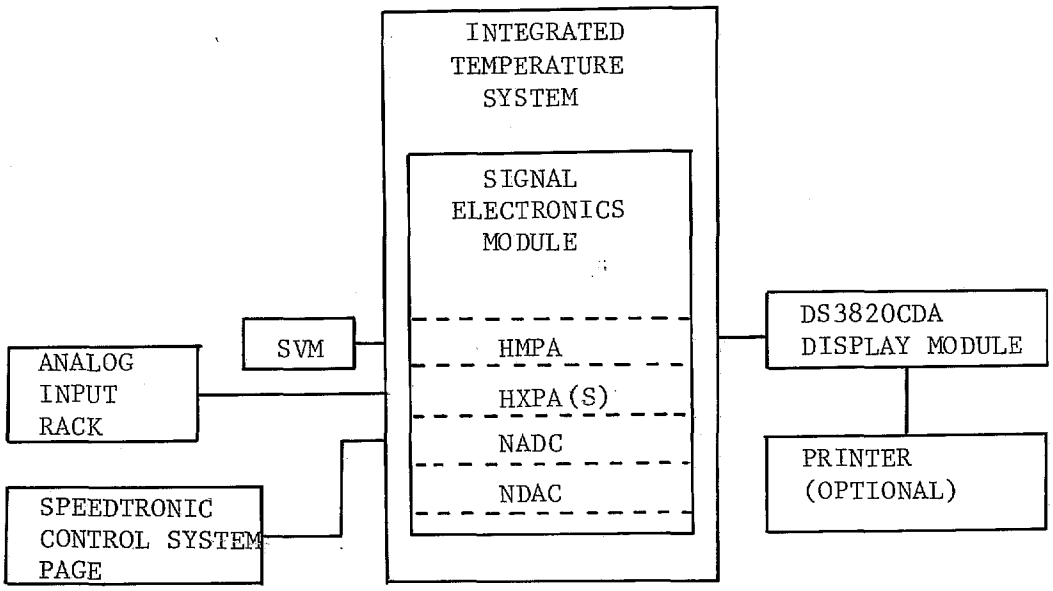


Figure 1-3. ITS System Test Block Diagram

NOTE

Panel markings and control names actually appearing on the equipment are shown in upper case letters throughout this publication.

SECTION 2
OPERATING INSTRUCTIONS

2-1 PURPOSE

The purpose of the operating instruction is to enable the user to interpret the test results obtained by use of the SVM and Display Module. The final result is to identify faulty ITS cards (not faulty components on those cards). However, the SVM is capable of providing some information below card level.

The following cards are tested by use of these instructions:

CARD LOCATION	NAME OF CARD
Signal Electronics Module	DS3800HMPA
	HXPA
	NADC
	NDAC
Analog Input Rack	IC3600QEAA
	AEXA
	ATCB
SPEEDTRONIC Page	IC3600AMIP
	AAIB
Display Module	DS3800HDDA

NOTE

When the SVM is connected to the SEM test receptacle, the SVM memory is used instead of the SEM (HMPA) memory (addresses ØØØØH thru Ø7FFH). Therefore, the ITS hardware executes the diagnostic programs rather than the application procedures.

2-2. CONNECTING SVM TO MODULE TEST RECEPTACLE.

CAUTION

IMPROPER USE OF THE SVM CAN RESULT IN DAMAGE TO THE SYSTEM UNDER TEST. BEFORE ATTACHING THE SVM TO A SPEEDTRONIC SYSTEM, THE FOLLOWING STEPS MUST BE PERFORMED.

a. STOP TURBINE

b. REMOVE POWER FROM AUXILIARY MOTORS AND DEVICES.

IF PRECAUTIONS DESCRIBED IN STEPS a AND b ARE NOT TAKEN, OPERATION OF AUXILIARY DEVICES CAN OCCUR. A TURBINE TRIP WILL OCCUR IF THE SVM IS CONNECTED WHILE THE TURBINE IS OPERATING.

To connect the SVM to a test receptacle, refer to Figure 2-1 and proceed as follows:

1. Rotate the SVM black actuator level fully counter-clockwise.

CAUTION

IN THE FOLLOWING STEP, INSURE PROPER ALIGNMENT OF SVM WITH TEST RECEPTACLE TO AVOID BENDING CONNECTOR PINS WHICH COULD RESULT IN DAMAGE TO THE EQUIPMENT. ESPECIALLY CHECK THAT NO RIBBON CABLES ARE ROUTED BETWEEN THE SVM AND THE SEM SHEETMETAL, AS THEY WILL CAUSE MISALIGNMENT AND POSSIBLE BENDING OF PINS.

2. Carefully engage rear end of SVM over test receptacle. It will engage with very little pressure. Any resistance means a misalignment problem exists.

3. Firmly hold SVM against test receptacle and rotate actuator lever fully clockwise (1/4 - turn) to vertical position.

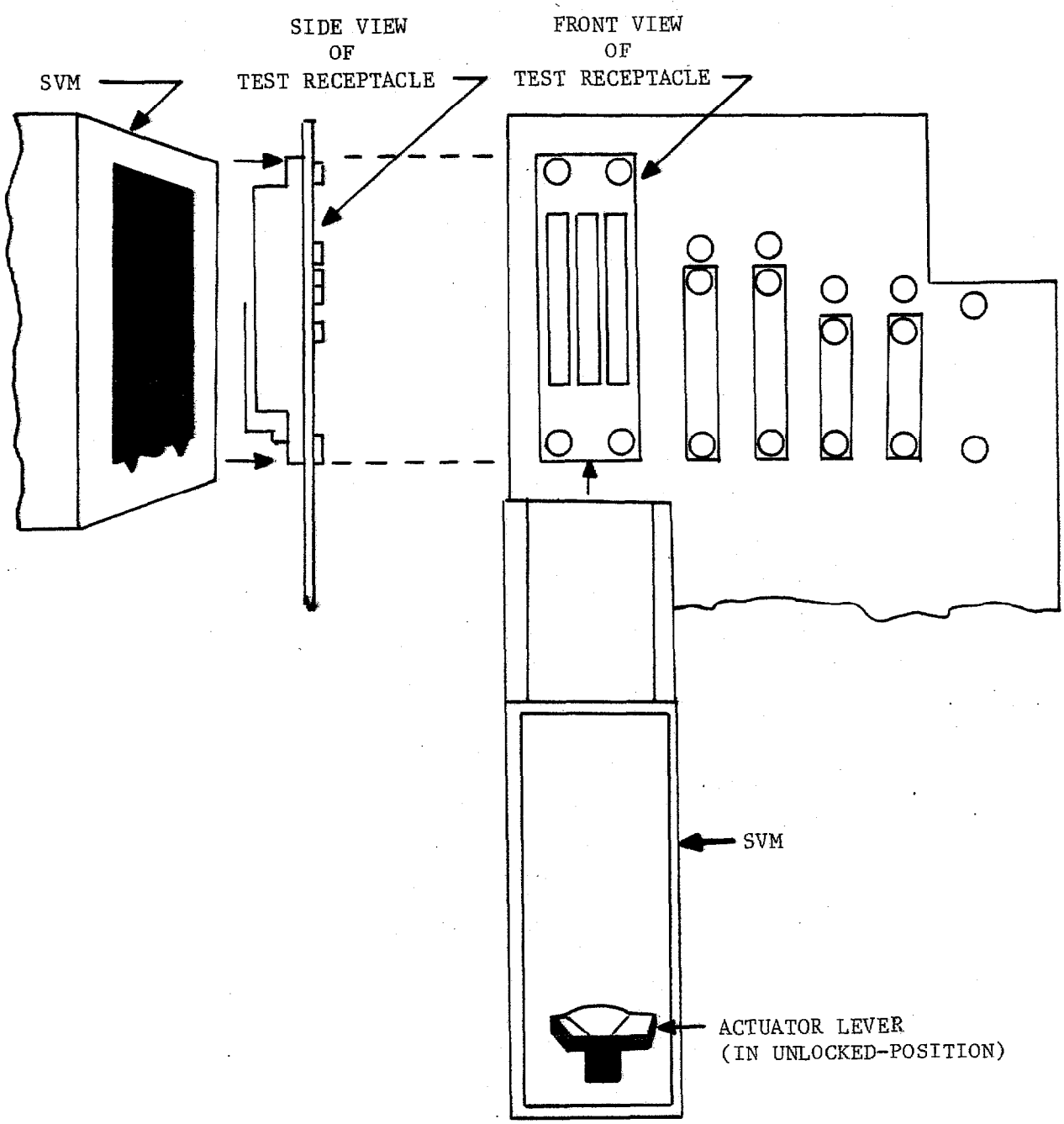


Figure 2-1 Engaging SVM With Test Receptacle

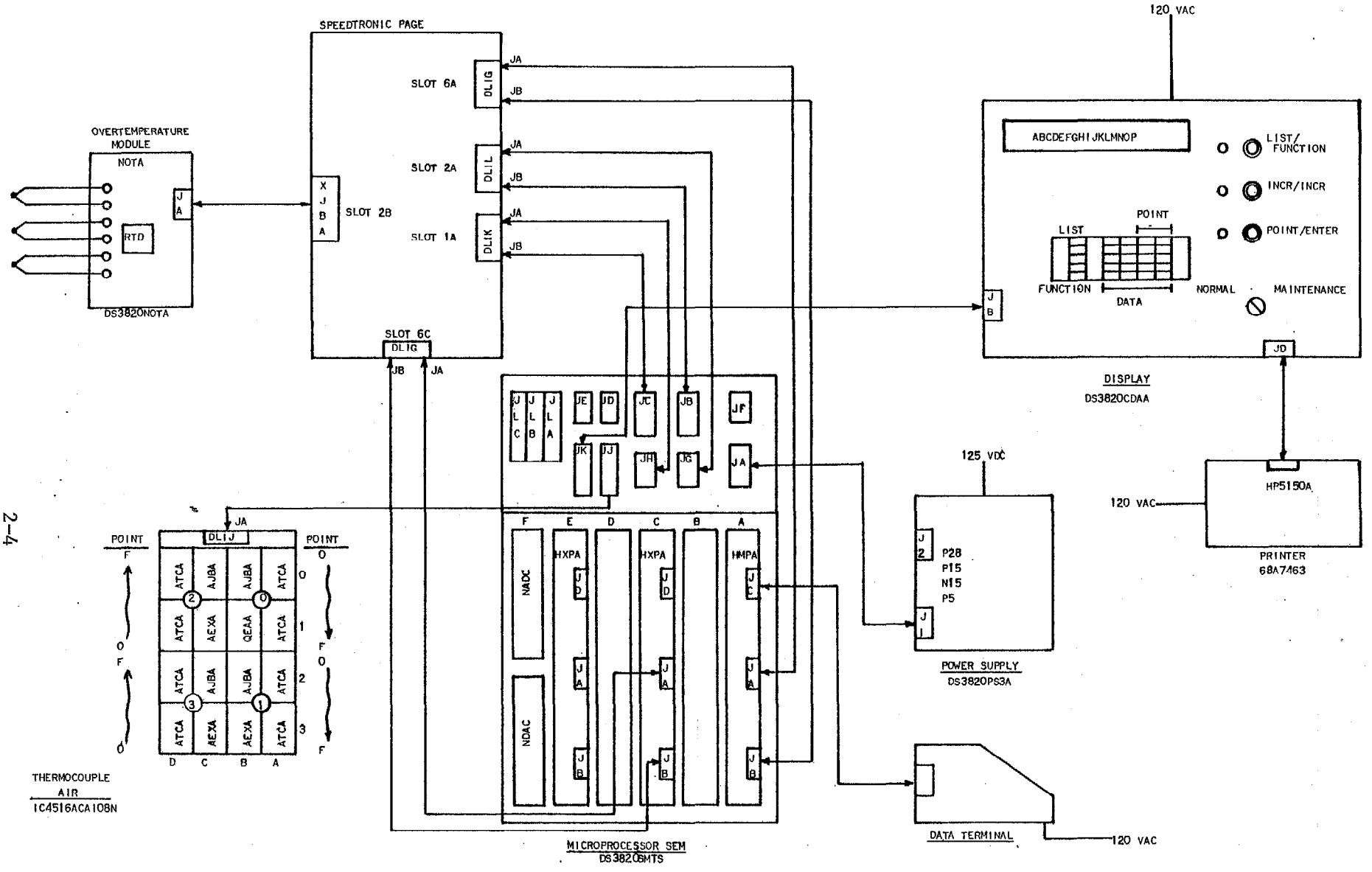


Figure 2-2 ITS System Test Connections

2-3 START-UP

1. Connect ITS system as shown in Figure 2-2, including printer if available.

NOTE

Actual equipment involved in the following tests is application dependent, therefore, the actual panel markings are unknown. For this reason, no attempt will be made to use actual control names in the following procedures.

2. Apply power to the signal electronics module (SEM), Display Module and connected peripheral equipment.

NOTE

The CPU status light codes are expressed as the Hexadecimal (BASE-16) equivalent of the binary number shown on the lights. A light ON = logic 1 and a light OFF = logic 0. See Figure 2-3. The status light code is normally a 2-digit alphabetic and/or numeric code with a suffix character "H" meaning hexadecimal.

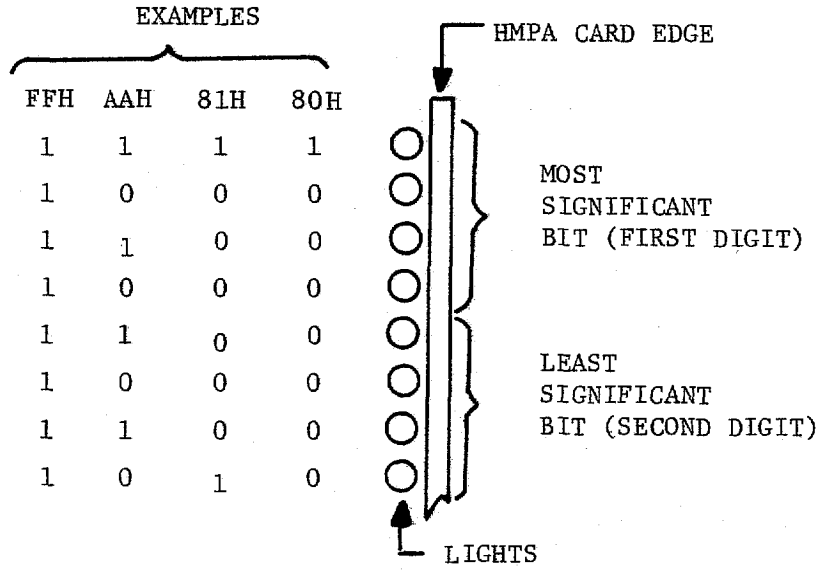
3. Observe CPU lights and depress HMPA toggle switch; lights should go out (display 00H), show 80H, followed by 81H, then go out.

4. The HMPA lights should now be off (00H) and the Display Module display should be blank.

NOTE

Controls specified in the following tests are located on the Display Module front panel unless otherwise specified.

The system is now prepared for desired test(s).



HEXADECIMAL	BINARY EQUIVALENT	HEXADECIMAL	BINARY EQUIVALENT
0	0 0 0 0	8	1 0 0 0
1	0 0 0 1	9	1 0 0 1
2	0 0 1 0	A	1 0 1 0
3	0 0 1 1	B	1 0 1 1
4	0 1 0 0	C	1 1 0 0
5	0 1 0 1	D	1 1 0 1
6	0 1 1 0	E	1 1 1 0
7	0 1 1 1	F	1 1 1 1
	8 4 2 1		8 4 2 1
	DECIMAL VALUE		DECIMAL VALUE

Figure 2-3. CPU Status Light Interpretation

2-4. TEST SELECTOR PROCEDURE. Any one or more of 14 different tests can be executed as follows:

1. Select desired test. This is accomplished by setting LIST thumbwheel switch to code corresponding to one of the following tests.

0 - Run Auto Test	7 - Stall Test
1 - CPU Test	8 - Parallel I/O Loopback Test
2 - Display Test	9 - Interrupt Test
3 - Printer Test	A.- NADC Test
4 - PROM Test	B.- NDAC Test
5 - RAM Test	C.- NDAC - NADC Loopback Test
6 - Timer Test	D.- USART (Teletype) Loopback Test

2. Depress POINT/ENTER pushbutton to start test.

2-5. CPU TEST

1. Set LIST thumbwheel switch to 1.

2. Test Parameters:

Exit: Runs to completion
Run Time: Approximately 1 second
Controls: None
Test OK message: 81H on CPU lights for about 1 second.
Error Messages:
FAH-CPU instruction failure (non-RAM).
FOH - CPU instruction failure (RAM).

F5H - Bad RAM somewhere in HMPA card ADDR ODFFF H to ODFF4 H (top 12 bytes).

EEH - CPU instruction failure causing a RST 7 (return or jump instruction).

Sequence of Operation: CPU status lights will display 00H, 80H, 81H, 00H, in sequence at about one second intervals.

2-6. DISPLAY TEST.

1. Set LIST thumbwheel switch to 2.

2. Test Parameters:

Exit: Part A - Change LIST thumbwheel switch;
Part B - KEYSWITCH set to NORMAL to return to Part A.

Controls: LIST thumbwheel switch - change to exit Part A and return to test selector procedure.

INCR pushbutton - push in Part A to go to single character mode.

ENTER pushbutton - push in Part A to return to scrolled mode from single step mode.

FUNCTION pushbutton - if keyswitch is in MAINTENANCE position, pushing FUNCTION will cause Part B to be entered.

Keyswitch - MAINTENANCE setting allows Part B to start when FUNCTION control is pushed; NORMAL setting returns to Part A from Part B.

Test OK Message: None (Operator must determine).

Error Message: None (operator must determine).

Run Time: About 50 seconds for a complete set of characters to scroll across display.

Sequence of Operation (See Table 2-1 for display test output):

a. Test starts by sending characters to the display. The complete character set is scrolled across the display changing once per second. If the INCR pushbutton is depressed, automatic scrolling stops and one new character is scrolled into the display each successive time INCR is pushed. Pushing ENTER returns the automatic scrolling.

b. Part B is entered by setting the keyswitch to MAINTENANCE and depressing FUNCTION pushbutton. The display echos the settings of the thumbwheel switches on the display:

L XHTO MAINT.
Where: L = LIST thumbwheel switch
X = THOUSANDS thumbwheel switch
H = HUNDREDS thumbwheel switch

TABLE 2-1

SELF VERIFICATION MODULE DISPLAY TEST OUTPUT

CHARACTER	DESCRIPTION
A thru Z	Alphabet
[	Left Square Bracket
~	Tilde ("Squiggle")
]	Right Square Bracket
{	Left Braces
}	Right Braces
	Blank
!	Exclaimation Point
"	Quote (Double)
#	Number Sign
$	Dollar Sign
%	Per Cent Sign
&	Ampersand
'	Apostrophe
(	Left Parenthesis
)	Right Parenthesis
*	Asterisk
+	Plus Sign
,	Comma
-	Hyphen (Minus Sign)
.	Period
/	Slash
0 thru 9	Numbers
:	Colon
;	Semi-Colon
<	Left Angle Bracket ("Less Than")
=	Equal Sign
>	Right Angle Bracket ("Greater Than")
?	Question Mark
@	"At" Sign

START AT THE TOP OF COLUMN AND READ DOWN.

T = TENS thumbwheel switch
O = ONES thumbwheel switch.

The message flashes on and off.

There is up to one second delay between changing a thumbwheel switch and display update. The pushbutton lights will come on if the corresponding pushbutton is depressed. There is up to one second delay between operation of a pushbutton and the corresponding light. If the keyswitch is returned to NORMAL, the test returns to Part A, where an exit to the test selector is possible.

2-7 PRINTER TEST

1. Verify that printer cable is connected to jack JD of Display Module. If cable is not connected, then turn power supply off, install cable, and turn power supply on. Turn printer on.
2. Set LIST thumbwheel switch to 3.
3. Test Parameters:

Exit: Change LIST thumbwheel switch.

Controls: LIST thumbwheel switch - change setting to exit and return to test selector.

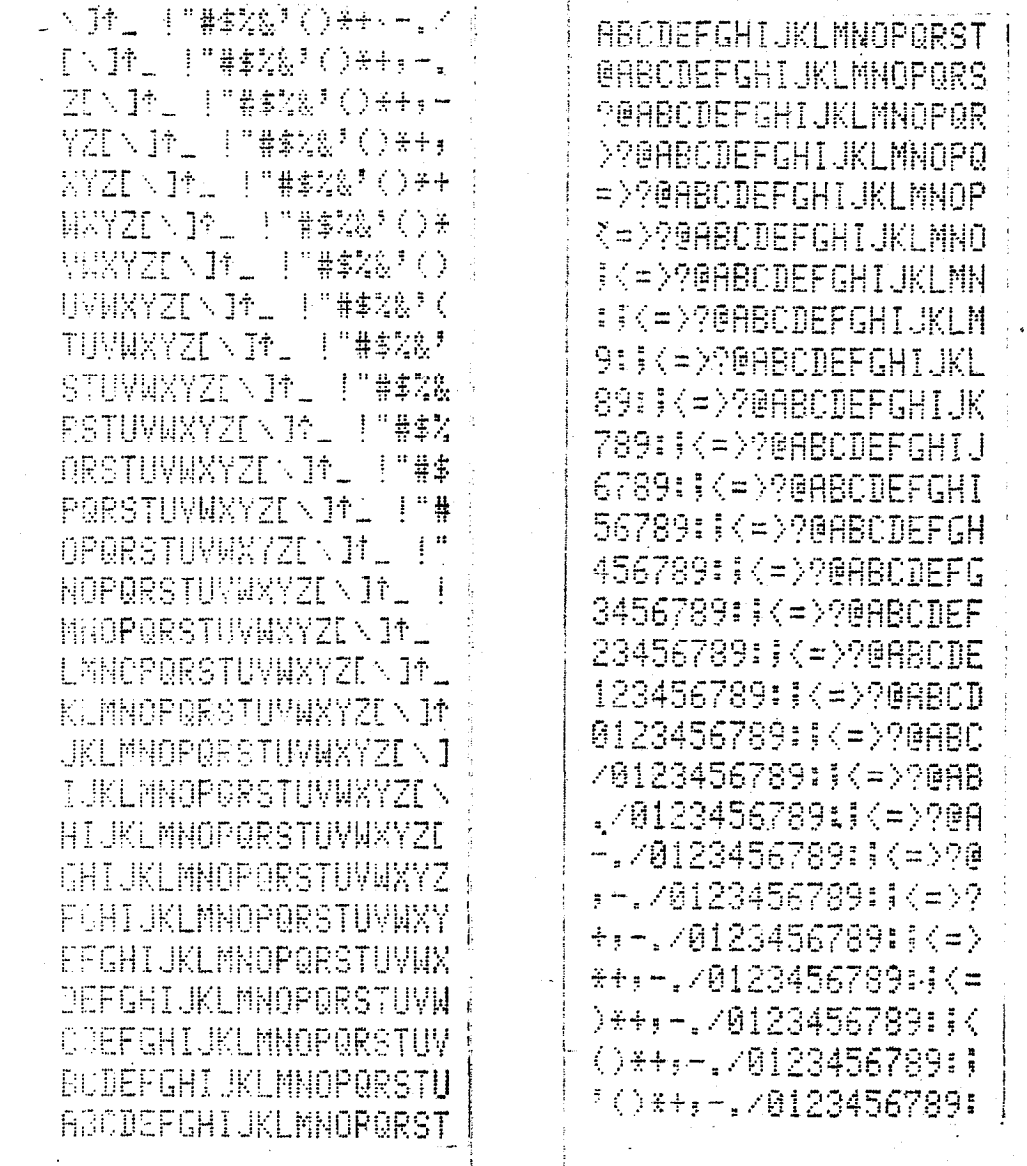
Test OK Message: None (determined by operator).

Error Messages: None (determined by operator).

Run Time: About 65 seconds for a complete set of characters.

Sequence of Operation: (See Table 2-2) This test outputs successive 20-character strings to the printer. The string "scrolls" one new character in and one character out each step, similar to display test.

Table 2-2. Self Verification Module Printer Test Output.



COLUMN 1 COLUMN 2

START AT THE BOTTOM OF COLUMN 1 AND READ UP TO THE TOP, THEN START AT THE BOTTOM OF COLUMN 2 AND READ UP.

2-8 PROM TEST.

1. Set LIST thumbwheel switch to 4.

2. Test Parameters:

Exit: Runs to completion (normal);
FUNCTION pushbutton (failure).

Controls: HUNDREDS thumbwheel switch-set to most significant digit of PROM count.

TENS thumbwheel switch - set to least significant digit of PROM count.

FUNCTION pushbutton - return to test selector after finding bad PROM.

INCR pushbutton - test next PROM after finding bad PROM.

ENTER pushbutton - ends "WAITING" mode and test reads the thumbwheel switch for PROM count.

Test OK Message:: END OF TEST #4 appears in display for 5 seconds.

Error Message: FAILED TEST #4 appears in display. Bad PROM # on CPU lights in Hex.

Run Time: About 0.1 second per PROM

Sequence of Operation: The word WAITING will appear on the display and 34H will appear on the CPU status lights. Count the number of PROMS in the system (normally four on HMPA and eight on each HXPA) and convert that number to Hex.

Example: 12 PROMS converts to OC Hex.

Set that Hex number into the HUNDREDS and TENS thumbwheel switches. The maximum number allowable is 28 (1CH). Larger numbers will not be accepted by the test. Push the ENTER pushbutton. If the number set on the thumbwheel switch is acceptable, the program will clear the display, turn off the lights and start at PROM #1 (start ADDR 0000H). The test calculates the checksum for the PROM and compares it to the checksum stored in the PROM. If the checksums match, the next PROM is tested and the process continues until the number of PROMs tested equals the number set in the thumbwheel switch. The CPU status lights will count up from zero showing the number of PROMs tested.

If a PROM fails the checksum test (missing PROM, erroneus checksum stored in PROM, or bad PROM), the FAILED TEST #4 message will be displayed and the CPU lights will indicate the Hex number corresponding to the failed PROM:

Hex	Card	Socket	Checksum ADDR*	
01	HMPA	U75	03FEH	
02	HMPA	U76	07FEH	If DS3820SVMA
03	HMPA	U77	0BFEH	is used, see
04	HMPA	U78	0FFEH	second
05	HXPA #1	U29	13FEH	paragraph
06	HXPA #1	U30	17FEH	after this table
07	HXPA #1	U31	1BFEH	
08	HXPA #1	U32	1FFEH	
09	HXPA #1	U33	23FEH	
0A	HXPA #1	U34	27FEH	
0B	HXPA #1	U35	2BFEH	
0C	HXPA #1	U36	2FFEH	
0D	HXPA #2	U29	33FEH	
0E	HXPA #2	U30	37FEH	
0F	HXPA #2	U31	3BFEH	
10	HXPA #2	U32	3FFEH	
11	HXPA #2	U33	43FEH	
12	HXPA #2	U34	47FEH	
13	HXPA #2	U35	4BFEH	
14	HXPA #2	U36	4FFEH	
15	HXPA #3	U29	53FEH	
16	HXPA #3	U30	57FEH	
17	HXPA #3	U31	5BFEH	
18	HXPA #3	U32	5FFEH	
19	HXPA #3	U33	63FEH	
1A	HXPA #3	U34	67FEH	
1B	HXPA #3	U35	6BFEH	
1C	HXPA #3	U36	6FFEH	

*Checksum stored as two bytes, LSB in LSA and MSB in LSA +1.

With the failure messgge on the display, one of two paths may be selected with the pushbuttons. The INCR pushbutton will cause the test to continue with the next PROM and the FUNCTION pushbutton will exit the test to the test selector.

When using the DS3820 SVMA, the self verification program is substituted for the program in the 4 PROMs on the HMPA card. The checksum test of the first 4K is performed on the SVM PROMs, not on the PROMS on the HMPA card. To test those 4 PROMs, remove them from the sockets on the HMPA card and either plug them into the next 4 unused sockets on the HXPA card or substitute them for any 4 other PROMs and run the test. The checksum test does not require the

PROM to be in any particular socket.

After the tests are finished, ensure that all PROMs are returned to the current sockets.

PROMs should not be removed unless their performance is suspect.

2-9. RAM TEST

1. Set LIST thumbwheel switch to 5.

2. Test Parameters:

Exit: Runs to completion (normal);
FUNCTION pushbutton (failure).

Controls:
ONES thumbwheel switch - set to number of K of RAM to be tested.

FUNCTION Pushbutton - return to test selector after finding bad RAM.

INCR Pushbutton - test next RAM after finding bad RAM.

ENTER Pushbutton - ends "waiting" mode and test reads the ONES thumbwheel switch for RAM count.

Test OK Message: END OF TEST #5 appears on display readout for 5 seconds.

Error Messages: BAD RAM AT ____H on display with the address of the bad byte in the blanks.

FAIL - STACK RAM appearing on display readout - Indicates bad RAM on the HMPA card.

BUS HOG - Attempt to test RAM not present in the system (usually means the thumbwheel switch was set improperly).

Run Time: About 60 seconds for each K of RAM tested.

Sequence of Operation: The test will first check a portion of RAM on the HMPA card to be used for a stack. Failure in this area will produce the message "FAIL-STACK RAM" indicating a bad RAM on the HMPA card between addresses DFE9 and DFFF. The only exit is accomplished by depressing FUNCTION pushbutton to return to the test selector. If that area tests OK, the test will wait with the "WAITING" message on the display and 35H on the CPU status lights. Set the ONES thumbwheel switch as follows for the RAM to be tested:

Amount of RAM	Thumbwheel Switch	Range of ADDR Tests
1K (HMPA)	1	DC00H-DFFFH
3K (HMPA and 1-HXPA)	3	D400H-DFFFH
5K (HMPA and 2-HXPA)	5	CC00H-DFFFH
7K (HMPA and 3-HXPA)	7	C400H-DFFFH

Note that setting the thumbwheel switch to too large a number will cause a bus hog (attempt to read/write non-existent device) which terminates the test and returns to the test selector via the CPU test.

Depress the ENTER pushbutton to cause the data to be read. If the number set on the ONES thumbwheel switch is 1, 3, 5, or 7, the test will continue. Any other setting will not be accepted and the test will continue to wait.

When the test starts, the display will clear and the lights will count the number of patterns that have been tested. The procedure will test each byte with 256 different patterns, and the lights will count up from 0 to FFH (256 counts) as the test progresses. If a byte fails any test, the display will show the message BAD RAM AT __H, with the blanks giving the address of the bad byte. The card containing the bad RAM can be determined as follows:

ADDR Between	Card
DC00-DFFFH	HMPA
D400-DBFFH	HXPA #1
CC00-D3FFH	HXPA #2
C400-CBFFH	HXPA #3

With the failure message appearing on the display, one of two paths may be selected using the pushbuttons. The INCR pushbutton will continue the test and the FUNCTION pushbutton will exit the test to the test selector. Since a bad RAM may not be found unless all patterns are tested, the test should be run to completion even after a bad RAM has been found.

2-10 TIMER TEST

1. Set LIST thumbwheel switch to 6.

2. Test Parameters:

   Exit: Runs to completion (normal); FUNCTION or INCR pushbutton (failure).

   Controls: FUNCTION pushbutton - return to test selector after test failure.

   INCR pushbutton - return to test selector after test failure.

   Test OK Message: END OF TEST #6 on display for 5 seconds.

   Error Message: FAILED TEST #6 on display.

   Run Time: Approximately one second.

   Sequence of Operation: The test checks timer 0 on the HMPA card. The results are either pass or fail.

2-11 STALL TEST

1. Set LIST thumbwheel switch to 7.

2. Test Parameters:

   Test OK Message: HMPA IMOK light goes out about one second after the test starts, remains off 4 seconds, then comes back on again after the test is complete.

   Error Message: None

   Run Time: Approximately 5 seconds.

2-12 PARALLEL I/O TEST

1. Verify that HMPA jack JA is connected to HMPA jack JB using test cable DS3815RCG1CL020X00. If not connected, then turn power supply off, install cable, and turn power supply on.
2. Set LIST thumbwheel switch to 8.
3. Test Parameters:

Exit: Runs to completion (normal)
FUNCTION pushbutton (failure)

Controls: FUNCTION pushbutton - return to test selector after failure.

INCR pushbutton - continue test after failure.

ENTER pushbutton - ends waiting mode and test starts.

Test OK Message: END TEST #8 message appears on display.

Error Message: FAILED TEST #8 message appears on display; status lights indicate test being performed.

Run Time: Approximately 25 seconds.

Sequence of Operation: The test will wait with the message "LINK I/O - HMPA" on display readout and 38H appears on lights until ENTER pushbutton is depressed. The test then begins. The CPU status lights indicate the progress of the test and provide information in the event of a failure:

Lights	Testing
A5	PI01A (HMPA)*
5A	PI01A (HMPA)*

*Unless a failure occurs, these will not be visible (duration about 20 microseconds).

Walking Bit **	PI01A

** A single light "on" is moved one bit at a time upward on the lights at about 1 second intervals.

55	PI02B(OUT)/PI02A(IN)
AA	PI02B(OUT)/PI02A(IN)
CC	PI02C
56	PI02C
A9	PI02C

If a failure occurs, the message "FAILED TEST #8" will appear on the display and the code of the failed test will appear on the CPU lights. One of the two paths may be selected using the pushbuttons. The FUNCTION pushbutton will terminate the test and return to the test selector. The INCR pushbutton will continue in the test. Since there are three areas being tested (PI01A, PI02A/PI02B, and PI02C), it is desirable to continue this test to its conclusion after a failure. As a point of information, the Display Test tests the PI01B and PI01C, the I/O ports used with the display.

2-13 INTERRUPT TEST

1. To prepare for test, preset keyswitch (See Controls section below).

2. Set LIST thumbwheel switch to 9.

3. Test Parameters:

Exit: Runs to completion (normal); FUNCTION pushbutton (failure); Halt Test - toggle RESTART switch on HMPA card.

Controls: Keyswitch - MAINTENANCE position will allow execution of a test of the "HALT" instruction. NORMAL position allows this portion of test to be skipped.

FUNCTION pushbutton - return to test selector after failure.

INCR pushbutton - continue in the test after failure.

Test OK Message: END TEST #9 appears on display for 5 seconds.

Error Message: FAILED TEST #9 appears on display; supplementary information appears on CPU lights.

Run Time: Approximately 15 seconds.

Sequence of Operation: The CPU lights are used to indicate the progress of the test and to provide data concerning the interrupt being tested:

Lights	Testing (HMPA Card-Only)
90	IRO (Bus Hog) - forces bus hog with interrupts masked. A failure indicates failure of the mask or the DI instruction.
91	Passed the first test.
90	IRO (Bus Hog) - forces bus hog with interrupts unmasked. A failure indicates that the bus hog failed to interrupt, or went to wrong service routine.
91	IR1 (clock) - checks clock interrupt timing and service routine used.
92	IR2 (analog input) - checks analog input timing and service routine used.

With the keyswitch set to NORMAL, the test is now complete.

If the keyswitch is set to MAINTENANCE:

93	HALT test - puts message "HALT TEST - RESTART" on display readout, waits 3 seconds, then halts. A failure indicates the HALT instruction did not work. The hardware restart must be used to restart the CPU after this portion of the test.

Since three interrupts are being tested, it is desirable to continue this test to its conclusion after a failure.

2-14 NADC TEST

1. To prepare for test, connect scope or meter to monitor ADDR lines.
2. Set LIST thumbwheel switch to A.
3. Test Parameters:

   Exit: Change LIST thumbwheel switch and depress ENTER or INCR pushbutton (normal),

   FUNCTION pushbutton (failure),

   FUNCTION or INCR pushbutton (interrupt failure).

Controls: FUNCTION pushbutton - return to test selector after failure.

INCR pushbutton - continue in test after failure, except that "failure to interrupt" (Test A-01) returns to test selector.

ENTER pushbutton -

(a) Ends initial waiting period and allows point to be read from thumbwheel switch input.

(b) Starts another set of readings if keyswitch is set to NORMAL.

Keyswitch - NORMAL requires pushing ENTER pushbutton for each set of readings. MAINTENANCE setting causes readings to be taken continuously.

LIST thumbwheel switch - Changing this switch setting results in exit to test selector.

Test OK Message: Determined by operator.

Error Message: FAILED TEST # A appears on display; supplementary information appears on CPU lights.

Run Time: Approximately 25 seconds for basic test.

Sequence of Operation: The test starts by raising one analog address line at a time high and leaving it there (no return to zero) for one second. The CPU lights indicate which line is high. Next, all address lines are taken high once per second, not returned to zero, for eight seconds. Thus, all lines should remain high eight seconds. The CPU status lights will indicate FF. Then the test takes all lines high once per second for eight seconds, return to zero. Thus, all lines should pulse high once per second. The CPU lights indicate 00. The test then checks the analog interrupt and the conversion time delay. A failure is indicated by FAILED TEST A on the display and the CPU status lights as follows:

Lights	Test Failed
01	Failure of interrupt
02	Short conversion time delay.
03	Long conversion time delay.

If the foregoing tests are successful, the test will wait with the message SET POINT ADDR on the display and 41 on the CPU lights until the ENTER pushbutton is depressed. It then reads the channel number from the TENS thumbwheel switch and the point number from the ONES thumbwheel switch and takes five inputs in succession from that point. The following results are displayed:

CP = RD1 RD5 SRD5D

where: C = channel #(0-7)

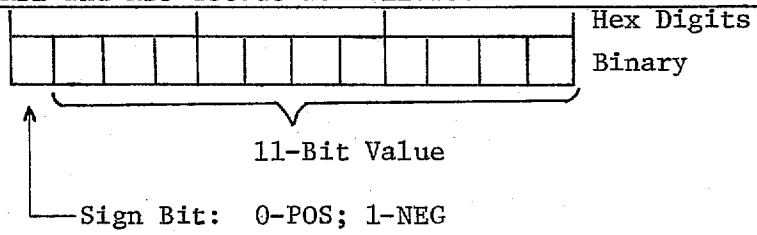
P = point # (0-F)

RD1 = hex input of first reading (000-FFF)

RD5 = hex input of fifth reading (000-FFF).

SRD5D = signed decimal value of fifth reading (+2047 to -2048).

RD1 and RD5 decode as follows:



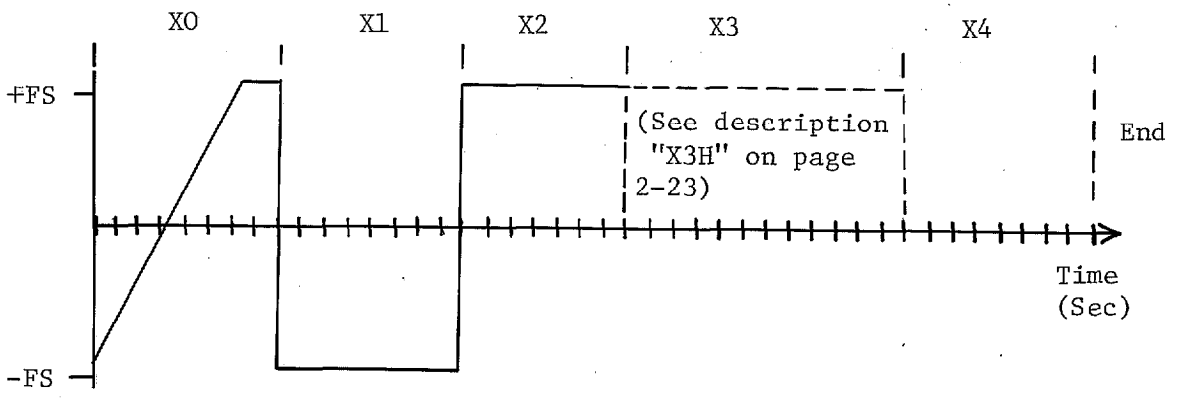
Thus, hex numbers greater than 7FF are negative input values in twos complement:

Analog	Digital (Hex)	Decimal
+Full Scale	7FF	+2047
	⋮	⋮
Zero +	001	+0001
Zero	000	+0000
Zero-	FFF	-0000
	⋮	⋮
-Full Scale	800	-2048

(Refer to Appendix 2 for volts-to-decimal, hex counts conversion).

Figure 2-4. Decoding Hex Values

If a chart recorder is used, the trace should appear as follows:



When the program finishes, the test exits to the test selector.

2-16. ANALOG (NDAC-NADC) LOOPBACK TEST:

1. To prepare for test, turn power supply off, connect NDAC output (SEM jack JC) to NADC output (SEM jack JC) to NADC input (SEM jack JJ) using special test cable 277A6581G1, and turn power supply on.

2. Set LIST thumbwheel switch to C.

3. Test Parameters:

Exit: a. Runs to completion, or
b. Change LIST thumbwheel switch.

Controls: LIST thumbwheel switch - changing setting of LIST thumbwheel will cause test to exit to test selector.

FUNCTION pushbutton - increment to next channel.

INCR pushbutton - continuous made: enter single-step mode. Single-step mode: make another step. Waiting mode: ends waiting mode.

ENTER pushbutton - single-step mode: return to to continuous mode.

Test OK Message: END OF TEST # CX at end of each channel where X is the channel number (0 to 3) for five seconds on display.

Error Message: OX OUT INP SDECM if |(Input-Output)|> five bits, where

X = Channel number
OUT = Hex value output*
IN = Hex value input*
S = Sign (+ or -) of input
DECM = Decimal value of input in counts (-2048 to +2047)

* See Figure 2-4.

Refer to Appendix 2 for volts-to-counts table.

Sequence of Operation: The test displays the message LINK UP D/A-A/D with 43 on the CPU lights and waits until the ENTER pushbutton is depressed. The CPU lights are used to monitor the progress of the test:

Lights**	Testing
X0	Finding average offset of input from output.
X1	Starts at -FS and does output, input, and compare of each point on a ramp to +FS. An error switches from continuous mode to single-step mode.
X2	Ramp +FS to -FS, checking each point. An error switches from continuous mode to single-step mode.

** "X" indicates channel under test (0 to 3)

At the end of each channel, the message END OF TEST # CX appears for five seconds and the test of the next channel begins.

Upon detection of an error, the test switches from continuous mode (ramp) to single-step mode. In this mode, an increment output-input-compare-display (IOICD) cycle is performed each time the INCR pushbutton is pushed. Each increment will change the output by 0.00488VDC. The ENTER pushbutton will cause the continuous mode to be re-entered.

If the FUNCTION pushbutton is depressed, the test of the current channel is terminated and the test of the next channel is started.

After all four channels (0 to 3) have been tested, the test exits to the test selector. If the LIST thumbwheel switch position is changed, the test exits to the test selector.

The CPU lights also show "channel" and "point" in Hex. With the keyswitch set to NORMAL position, the thumbwheel switch will be read and a set of readings taken and displayed each time the ENTER pushbutton is depressed.

If the keyswitch is set to the MAINTENANCE position, this test will not wait for the ENTER pushbutton but will read the thumbwheel switch and take a new set of readings once each second. To exit from this automatic cycle mode, change the LIST thumbwheel switch setting.

The test will exit to the test selector after taking one more reading. If in the manual cycle mode, change LIST thumbwheel switch setting and depress ENTER pushbutton to take one more reading.

NOTE

The channel number (TENS thumbwheel switch) ranges from 0 to 7. Setting the TENS thumbwheel switch greater than 7 will produce a channel number as follows:

Thumbwheel Switch Setting	Channel
8	0
9	1
A	2
B	3
C	4
D	5
E	6
F	7

Point numbers on the ONES thumbwheel switch range from 0 to F.

2-15. NDAC TEST

1. To prepare for test, connect a voltmeter or chart recorder to the analog output channel to be tested.

2. Set LIST thumbwheel switch to B.

3. Test Parameters:

   Exit: Runs to completion and exits to test selector.

   Controls: ENTER pushbutton - ends waiting mode and starts output test.

TENS thumbwheel switch - selects channel to be tested (0 to 3).

Test OK Message: Determined by operator.

Error Message: BUS HOG - missing card.

Run Time: Approximately 26 seconds per channel.

Sequence of Operation: This test will verify that the NDAC card is present and responding. If no bus hog occurs, the message SET POINT ADDR is displayed with 42 on the CPU lights. The test then waits until the ENTER pushbutton is depressed. When this occurs, the test reads the TENS thumbwheel switch for the channel number to be tested.

Channel #	Thumbwheel Switch Settings
0	0, 4, 8, C
1	1, 5, 9, D
2	2, 6, A, E
3	3, 7, B, F

The progress of the test is indicated on the CPU lights. The operator must monitor the output and decide if the card passes:

Lights*	Output
X0H	Ramp from -FS (-10VDC) to +FS (+10VDC) and hold at +FS one second. (4.7 seconds total).
X1H	-FS (-10VDC) for five seconds.
X2H	+FS (+10 VDC) for five seconds.
X3H	The D/A is "frozen" by forcing a stall. Two seconds later a command for -FS (-10VDC) is made for five seconds. The output behavior depends on the condition of two backplane straps on the NADC card. The signals ODZR1/2 and ODZR3/4, pins 33 and 34 respectively, control channels 1 and 2 (X = Ø,1) and channels 3 and 4 (X = 2, 3) respectively. If the pin is open, the output will remain at +FS. If the Pin is connected to common, the output will be zero (forces FFF_H on input to D/A chip, which is -1 decimal = -0.005 VDC output).
X4H	0 (0VDC) for five seconds.

*X indicates the channel under test (0 to 3).

2-17 USART TEST.

1. To prepare for test, install test plug 277A6580G1 in the HMPA serial I/O socket JC prior to entering the test from the test selector.

2. Set LIST thumbwheel switch to D.

3. Test Parameters:

   Exit: Runs to completion (normal).

   FUNCTION pushbutton (failure)

   Controls: None (normal).

   FUNCTION pushbutton - return to test selector (failure).

   Test OK Message: END OF TEST #D appears on display for 5 seconds.

   Error Message: FAILED TEST # DXX where XX = error number. Any error message indicates failure of the USART function on the HMPA card. The specific error number is not significant in deciding whether the card is good or bad. The error number(s) should be reported with the failed card for factory repair purposes.

   Run Time: Approximately 0.1 seconds.

   Sequence of Operation: The test checks a number of features of the USART and accumulates error messages. After all tests are run, the test exits to the test selector if no errors have occurred. If errors are observed, the first error message is displayed. If the INCR pushbutton is depressed, the next error number is displayed. If the function PB is depressed, the test exits to the test selector. If INCR is pushed after the last error is displayed, the test runs again.

2-18. RUN AUTO TEST:

1. To prepare for test, set the following controls to indicated positions:

Control	Position
LIST thumbwheel switch	0 (selects AUTO test).
TENS thumbwheel switch	Least significant digit of PROM count.
HUNDREDS thumbwheel switch	Most significant digit of PROM count.
ONES thumbwheel switch	Set to number of K of RAM to be tested.
Keyswitch	NORMAL

Connect HMPA jack JA to HMPA jack JB with test cable DS3815RG1CL020X00 for parallel I/O test.

Connect SEM jack JC to SEM jack JJ with test cable 277A6581G1 for analog loopback test.

2. Test Parameters:

Exit: Change LIST thumbwheel switch (normal).

Refer to test instructions for test that failed (failure).

Controls: See test instructions for test(s) that failed.

Test OK Message: See specific tests.

Run Time: The run time of the AUTO test is the sum of the running time of all tests executed.

Sequence of Operation:

The AUTO test runs the following tests in sequence as long as the ENTER pushbutton is depressed. Each test runs as described in the foregoing test instructions. The AUTO test simply selects the tests to run and the sequence in which they run, shown below:

CPU (Test #1)
PROM (Test #4)
RAM (Test #5)
TIMER (Test #6)
STALL (Test #7)
PARALLEL I/O (Test #8)
INTERRUPT (Test #9)
ANALOG LOOPBACK (Test #C).

A failure will stop the test cycle and retain the error indication. Refer to the specific test information for details.

2-19 PRINTER:

If a printer is connected to the system (connector JD of display module) and turned on, all Test OK and Error messages will come out on the printer as well as the display. The format for the End of Test and Failed Test # messages is slightly different. The tape reads from the bottom up, and the test # will be printed on the line above the message.

2-20 ANALOG INPUT INTERFACE

The analog input interface has two functions:

(1) To provide signal conditioning for the conversion of various signal levels to a standard range of -10VDC to 10VDC for input to the NADC card.
(2) To multiplex multiple points into a single NADC channel.

The interface is located in two areas and connects with ribbon cable to the SEM. The Analog Input Rack (AIR) is used primarily for thermocouple inputs (millivolts). The SPEEDTRONIC control system page contains IC3600AMIP cards for high level inputs (-10VDC to +10VDC range).

The systems elementary must be used to identify the channel and point used for a specific signal. A typical system is used for this discussion. Use the system elementary for a particular installation to identify channel and point numbers.

The self-verification module test "A" (NADC test) is used to scan the channel and point under consideration and display the results of the analog-to-digital conversion on the display. Refer to the NADC test instructions for specific test parameters. Only interpretation of the readings is covered in this paragraph.

Inputs Using IC3606ATCA/B, IC3600QEAA, IC3600AEXA Cards - Type K Thermocouple.

(1) Internal Resistance Temperature Detector (RTD) Sensors.

Point "0" on each ATCA/B card is an RTD used to measure the temperature of the junction between the type K lead wires and the card wiring. Since two ATCA/B's may be connected to each channel, and strapped to respond to either the low 8 or the high 8-point addresses by the NADC, the RTD is addressed at X0H or X8H, where X is the channel #, depending upon which ATCA/B card is being tested. The temperature of the RTD must be known to interpret the readings. Measure the temperature by inserting a thermometer in the hole provided in the terminal block of the ATCA/B under test. Allow the thermometer reading to stabilize (+/- 0.2 deg. F.) before taking the reading. Use test A (NADC test) to read the RTD signal. The relationship between the ATCA/B temperature and the RTD signal is given in Appendix 3. The table may be linearly extrapolated for ambient temperatures not listed. See paragraph (3) for interpretation of incorrect readings.

(2) Thermocouple Inputs.

Use a millivolt source to set test input voltages. The relationship between input millivolts and the display reading in decimal counts is:

Counts - MV X 74.023 $\pm$ 5 Counts

The millivolt source should be adjusted +/-0.005 MVDC.

Three test points are sufficient:

MV	Decimal	Hex
+24.902	1843	733H
0.000	0	0H
-24.902	-1843	8CDH

The ambient temperature of the ATCA/B does not influence the conversion. To fully test the multiplexing hardware, all other TC inputs should be shorted, and the input applied to a single point. That point should read correctly, and all other TC's should read 0. See paragraph (3) to interpret incorrect readings.

(3) If incorrect readings are obtained do not draw the conclusion that the channel/point, which reads incorrectly, is defective. The pattern of the readings must be analyzed to locate the defective portion of the system as follows:

Pattern	Probable Cause
a. One point reads incorrectly.	a. Defective point-replace ATCA/B.
b. All points except one on a single channel, read low.	b. The point reading correctly is defective-replace that ATCA/B.
c. One TC channel reading correctly, all other TC channels read low.	c. The AEXA or QEAA associated with the channel reading correctly is defective.
d. In a system with one QEAA, all TC channels read incorrectly.	d. Either the QEAA or the NADC is defective. If the QEAA output is connected to NADC channels 0, 2, or 3, test C may be used to determine if the NADC is at fault.
e. In a system with more than one QEAA, all TC channels read incorrectly.	e. The NADC is defective.

Pattern	Probable Cause
f. In a system with more than one QEAA, all TC channels feeding one QEAA read incorrectly.	The QEAA is defective.

SPEEDTONIC Page High-Level Interface Using IC3600AMIP

Test A is used to select a point and read its value. The relationship between the input voltage and the decimal counts displayed is:

Counts = V X 204.8 $\pm$ 8 Counts

The input voltage should be set +/- .01VDC.

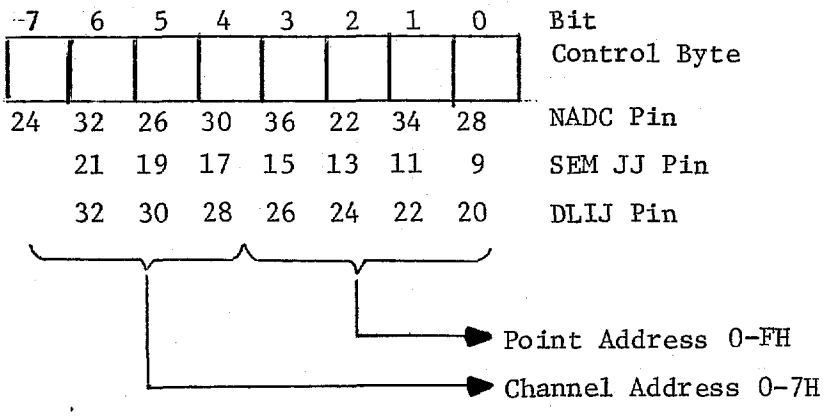
Typical Test Values:

Volts	Decimal Counts	Hex Counts
+9.00	1843	733H
+8.00	1638	666H
+5.00	1024	400H
0.00	0	0H
-7.00	-1434	59AH

The AMIP card provides only multiplexing. If a single point reads incorrectly, replace the AMIP. If all points read incorrectly, either the AMIP or the NADC may be at fault.

Point Selection

The control signals that select a single point in the multiplex tree are generated by the NADC in response to a control byte written to the card by the HXPA card. This control word is decoded as follows:



The signal flow is as follows:

The channel address selects one of the NADC input channels, and also selects one of the QEAA or AEXA cards if a TC channel is being addressed. The strapping of the card determines the channel address to which it will respond. The QEAA or AEXA card selected generates a signal which enables both ATCA/B cards connected to the selected channel. Bit 3, in turn, enables one of the ATCA/B cards, depending on how the card is strapped. Bits 0-2 select one of the 8 points on the ATCA/B card.

The AMIP card(s) will respond to a control byte by enabling the input point corresponding to the point address (0-F). The point input will be applied to the NADC but will only be read if that NADC channel has been selected.

Incorrect readings may result from problems in address control signals. If the observed failure pattern does not change when the suspect cards are replaced, the addressing may be at fault. Use an oscilloscope to "Read" the address lines and verify that the state of the address lines corresponds to the channel and point selected. The NADC address outputs are 1 = HIGH (normal sense).

APPENDIX 1
MESSAGE INTERPRETATION

The lights are used to pass three types of information:

(1) Progress of Test: The sequence of operation section of each test describes the action of the lights during the test. The patterns of the lights during normal operation of a test are not necessarily unique to that test.

(2) ID of Test Requesting Operator Input: Some of the tests pause until the operator takes an action to cause the test to continue. The actions are described in detail in the "sequence of operation" section of the test. The messages are summarized in table 1-A.

(3) Failure Messages: The lights provide supplementary information when the display indicates "Failed Test # ___", where the test number is in the blank. These messages are summarized in table 1-B.

(4) "BUS HOG" Message: An attempt to address something failed (no response from the device addressed). May be caused by a missing card, bad card, or backplane wiring failure.

TABLE 1-A

DISPLAY MESSAGE	LIGHTS	CONDITION
None	00	Waiting in Test Selector
"WAITING"	34	Waiting in PROM Test #4
"WAITING"	35	Waiting in RAM Test #5
"LINK I/O-HMPA"	38	Waiting in PARALLEL I/O Test #8
"SET POINT ADDR"	41	Waiting in NADC Test #A
"SET POINT ADDR"	42	Waiting in NDAC Test #B
"LINK-UP D/A-A/D"	43	Waiting in NADC-NDAC Loopback Test #C

TABLE 1-B

FAILURE MESSAGES

LIGHTS	INTERPRETATION AND CAUSES
01	NADC Failure to Interrupt - NADC Failure - HMPA Failure - Backplane Wiring
02	NADC Short Converstion Time Delay - NADC Failure - HMPA Clock Failure
03	NADC Long Conversion Time Delay - NADC Failure - HMPA Clock Failure
55	HMPA I/O Port 2 Failure
56	HMPA I/O Port 2 Failure
5A	HMPA I/O Port 1 Failure
90	HMPA Interrupt Mask OK IRO (Bus Hog) Failure
91	HMPA Interrupt IR1 (Clock) Failure
92	HMPA Interrupt IR2 (NADC) Failure
A5	HMPA I/O Port 1 Failure
A9	HMPA I/O Port 2 Failure
AA	HMPA I/O Port 2 Failure
CC	HMPA I/O Port 2 Failure
EE	HMPA CPU Failure (Jump or Call)
FO	HMPA CPU Failure (RAM Dependent Instruction)
F5	HMPA Stack RAM Bad (DFF3-DFFF)
FA	HMPA CPU Failure (Non-RAM Dependent Instruction)

APPENDIX 2

VOLTS-DECIMAL-HEX CONVERSION

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
9.995	2047	7FF	9.814	2010	7DA
9.990	2046	7FE	9.810	2009	7D9
9.985	2045	7FD	9.805	2008	7D8
9.980	2044	7FC	9.800	2007	7D7
9.976	2043	7FB	9.795	2006	7D6
9.971	2042	7FA	9.790	2005	7D5
9.966	2041	7F9	9.785	2004	7D4
			9.780	2003	7D3
9.961	2040	7F8	9.775	2002	7D2
9.956	2039	7F7	9.771	2001	7D1
9.951	2038	7F6			
9.946	2037	7F5	9.766	2000	7D0
9.941	2036	7F4	9.761	1999	7CF
9.937	2035	7F3	9.756	1998	7CE
9.932	2034	7F2	9.751	1997	7CD
9.927	2033	7F1	9.746	1996	7CC
9.922	2032	7F0	9.741	1995	7CB
9.917	2031	7EF	9.736	1994	7CA
			9.731	1993	7C9
9.912	2030	7EE	9.727	1992	7C8
9.907	2029	7ED	9.722	1991	7C7
9.902	2028	7EC			
9.897	2027	7EB	9.717	1990	7C6
9.893	2026	7EA	9.712	1989	7C5
9.888	2025	7E9	9.707	1988	7C4
9.883	2024	7E8	9.702	1987	7C3
9.878	2023	7E7	9.697	1986	7C2
9.873	2022	7E6	9.692	1985	7C1
9.868	2021	7E5	9.687	1984	7C0
			9.683	1983	7BF
9.863	2020	7E4	9.678	1982	7BE
9.858	2019	7E3	9.673	1981	7BD
9.854	2018	7E2			
9.849	2017	7E1	9.668	1980	7BC
9.844	2016	7E0	9.663	1979	7BB
9.839	2015	7DF	9.658	1978	7BA
9.834	2014	7DE	9.653	1977	7B9
9.829	2013	7DD	9.648	1976	7B8
9.824	2012	7DC	9.644	1975	7B7
9.819	2011	7DB	9.639	1974	7B6
			9.634	1973	7B5
			9.629	1972	7B4
			9.624	1971	7B3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
9.619	1970	7B2	9.424	1930	78A
9.614	1969	7B1	9.419	1929	789
9.609	1968	7B0	9.414	1928	788
9.604	1967	7AF	9.409	1927	787
9.600	1966	7AE	9.404	1926	786
9.595	1965	7AD	9.399	1925	785
9.590	1964	7AC	9.395	1924	784
9.585	1963	7AB	9.390	1923	783
9.580	1962	7AA	9.385	1922	782
9.575	1961	7A9	9.380	1921	781
9.570	1960	7A8	9.375	1920	780
9.565	1959	7A7	9.370	1919	77F
9.561	1958	7A6	9.365	1918	77E
9.556	1957	7A5	9.360	1917	77D
9.551	1956	7A4	9.355	1916	77C
9.546	1955	7A3	9.351	1915	77B
9.541	1954	7A2	9.346	1914	77A
9.536	1953	7A1	9.341	1913	779
9.531	1952	7A0	9.336	1912	778
9.526	1951	79F	9.331	1911	777
9.521	1950	79E	9.326	1910	776
9.517	1949	79D	9.321	1909	775
9.512	1948	79C	9.316	1908	774
9.507	1947	79B	9.312	1907	773
9.502	1946	79A	9.307	1906	772
9.497	1945	799	9.302	1905	771
9.492	1944	798	9.297	1904	770
9.487	1943	797	9.292	1903	76F
9.482	1942	796	9.287	1902	76E
9.478	1941	795	9.282	1901	76D
9.473	1940	794	9.277	1900	76C
9.468	1939	793	9.272	1899	76B
9.463	1938	792	9.268	1898	76A
9.458	1937	791	9.263	1897	769
9.453	1936	790	9.258	1896	768
9.448	1935	78F	9.253	1895	767
9.443	1934	78E	9.248	1894	766
9.438	1933	78D	9.243	1893	765
9.434	1932	78C	9.238	1892	764
9.429	1931	78B	9.233	1891	763

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
9.229	1890	762	9.033	1850	73A
9.224	1889	761	9.028	1849	739
9.219	1888	760	9.023	1848	738
9.214	1887	75F	9.019	1847	737
9.209	1886	75E	9.014	1846	736
9.204	1885	75D	9.009	1845	735
9.199	1884	75C	9.004	1844	734
9.194	1883	75B	8.999	1843	733
9.189	1882	75A	8.994	1842	732
9.185	1881	759	8.989	1841	731
9.180	1880	758	8.984	1840	730
9.175	1879	757	8.979	1839	72F
9.170	1878	756	8.975	1838	72E
9.165	1877	755	8.970	1837	72D
9.160	1876	754	8.965	1836	72C
9.155	1875	753	8.960	1835	72B
9.150	1874	752	8.955	1834	72A
9.146	1873	751	8.950	1833	729
9.141	1872	750	8.945	1832	728
9.136	1871	74F	8.940	1831	727
9.131	1870	74E	8.936	1830	726
9.126	1869	74D	8.931	1829	725
9.121	1868	74C	8.926	1828	724
9.116	1867	74B	8.921	1827	723
9.111	1866	74A	8.916	1826	722
9.106	1865	749	8.911	1825	721
9.102	1864	748	8.906	1824	720
9.097	1863	747	8.901	1823	71F
9.092	1862	746	8.896	1822	71E
9.087	1861	745	8.892	1821	71D
9.082	1860	744	8.887	1820	71C
9.077	1859	743	8.882	1819	71B
9.072	1858	742	8.877	1818	71A
9.067	1857	741	8.872	1817	719
9.062	1856	740	8.867	1816	718
9.058	1855	73F	8.862	1815	717
9.053	1854	73E	8.857	1814	716
9.048	1853	73D	8.853	1813	715
9.043	1852	73C	8.848	1812	714
9.038	1851	73B	8.843	1811	713

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
8.838	1810	712	8.643	1770	6EA
8.833	1809	711	8.638	1769	6E9
8.828	1808	710	8.633	1768	6E8
8.823	1807	70F	8.628	1767	6E7
8.818	1806	70E	8.623	1766	6E6
8.813	1805	70D	8.618	1765	6E5
8.809	1804	70C	8.613	1764	6E4
8.804	1803	70B	8.608	1763	6E3
8.799	1802	70A	8.604	1762	6E2
8.794	1801	709	8.599	1761	6E1
8.789	1800	708	8.594	1760	6E0
8.784	1799	707	8.589	1759	6DF
8.779	1798	706	8.584	1758	6DE
8.774	1797	705	8.579	1757	6DD
8.770	1796	704	8.574	1756	6DC
8.765	1795	703	8.569	1755	6DB
8.760	1794	702	8.564	1754	6DA
8.755	1793	701	8.560	1753	6D9
8.750	1792	700	8.555	1752	6D8
8.745	1791	6FF	8.550	1751	6D7
8.740	1790	6FE	8.545	1750	6D6
8.735	1789	6FD	8.540	1749	6D5
8.730	1788	6FC	8.535	1748	6D4
8.726	1787	6FB	8.530	1747	6D3
8.721	1786	6FA	8.525	1746	6D2
8.716	1785	6F9	8.521	1745	6D1
8.711	1784	6F8	8.516	1744	6D0
8.706	1783	6F7	8.511	1743	6CF
8.701	1782	6F6	8.506	1742	6CE
8.696	1781	6F5	8.501	1741	6CD
8.691	1780	6F4	8.496	1740	6CC
8.687	1779	6F3	8.491	1739	6CB
8.682	1778	6F2	8.486	1738	6CA
8.677	1777	6F1	8.481	1737	6C9
8.672	1776	6F0	8.477	1736	6C8
8.667	1775	6EF	8.472	1735	6C7
8.662	1774	6EE	8.467	1734	6C6
8.657	1773	6ED	8.462	1733	6C5
8.652	1772	6EC	8.457	1732	6C4
8.647	1771	6EB	8.452	1731	6C3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
8.447	1730	6C2	8.252	1690	69A
8.442	1729	6C1	8.247	1689	699
8.437	1728	6C0	8.242	1688	698
8.433	1727	6BF	8.237	1687	697
8.428	1726	6BE	8.232	1686	696
8.423	1725	6BD	8.228	1685	695
8.418	1724	6BC	8.223	1684	694
8.413	1723	6BB	8.218	1683	693
8.408	1722	6BA	8.213	1682	692
8.403	1721	6B9	8.208	1681	691
8.398	1720	6B8	8.203	1680	690
8.394	1719	6B7	8.198	1679	68F
8.389	1718	6B6	8.193	1678	68E
8.384	1717	6B5	8.188	1677	68D
8.379	1716	6B4	8.184	1676	68C
8.374	1715	6B3	8.179	1675	68B
8.369	1714	6B2	8.174	1674	68A
8.364	1713	6B1	8.169	1673	689
8.359	1712	6B0	8.164	1672	688
8.354	1711	6AF	8.159	1671	687
8.350	1710	6AE	8.154	1670	686
8.345	1709	6AD	8.149	1669	685
8.340	1708	6AC	8.145	1668	684
8.335	1707	6AB	8.140	1667	683
8.330	1706	6AA	8.135	1666	682
8.325	1705	6A9	8.130	1665	681
8.320	1704	6A8	8.125	1664	680
8.315	1703	6A7	8.120	1663	67F
8.311	1702	6A6	8.115	1662	67E
8.306	1701	6A5	8.110	1661	67D
8.301	1700	6A4	8.105	1660	67C
8.296	1699	6A3	8.101	1659	67B
8.291	1698	6A2	8.096	1658	67A
8.286	1697	6A1	8.091	1657	679
8.281	1696	6A0	8.086	1656	678
8.276	1695	69F	8.081	1655	677
8.271	1694	69E	8.076	1654	676
8.267	1693	69D	8.071	1653	675
8.262	1692	69C	8.066	1652	674
8.257	1691	69B	8.062	1651	673

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
7.666	1570	622	7.471	1530	5FA
7.661	1569	621	7.466	1529	5F9
7.656	1568	620	7.461	1528	5F8
7.651	1567	61F	7.456	1527	5F7
7.646	1566	61E	7.451	1526	5F6
7.642	1565	61D	7.446	1525	5F5
7.637	1564	61C	7.441	1524	5F4
7.632	1563	61B	7.437	1523	5F3
7.627	1562	61A	7.432	1522	5F2
7.622	1561	619	7.427	1521	5F1
7.617	1560	618	7.422	1520	5F0
7.612	1559	617	7.417	1519	5EF
7.607	1558	616	7.412	1518	5EE
7.603	1557	615	7.407	1517	5ED
7.598	1556	614	7.402	1516	5EC
7.593	1555	613	7.397	1515	5EB
7.588	1554	612	7.393	1514	5EA
7.583	1553	611	7.388	1513	5E9
7.578	1552	610	7.383	1512	5E8
7.573	1551	60F	7.378	1511	5E7
7.568	1550	60E	7.373	1510	5E6
7.563	1549	60D	7.368	1509	5E5
7.559	1548	60C	7.363	1508	5E4
7.554	1547	60B	7.358	1507	5E3
7.549	1546	60A	7.354	1506	5E2
7.544	1545	609	7.349	1505	5E1
7.539	1544	608	7.344	1504	5E0
7.534	1543	607	7.339	1503	5DF
7.529	1542	606	7.334	1502	5DE
7.524	1541	605	7.329	1501	5DD
7.520	1540	604	7.324	1500	5DC
7.515	1539	603	7.319	1499	5DB
7.510	1538	602	7.314	1498	5DA
7.505	1537	601	7.310	1497	5D9
7.500	1536	600	7.305	1496	5D8
7.495	1535	5FF	7.300	1495	5D7
7.490	1534	5FE	7.295	1494	5D6
7.485	1533	5FD	7.290	1493	5D5
7.480	1532	5FC	7.285	1492	5D4
7.476	1531	5FB	7.280	1491	5D3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
7.275	1490	5D2	7.080	1450	5AA
7.271	1489	5D1	7.075	1449	5A9
7.266	1488	5D0	7.070	1448	5A8
7.261	1487	5CF	7.065	1447	5A7
7.256	1486	5CE	7.061	1446	5A6
7.251	1485	5CD	7.056	1445	5A5
7.246	1484	5CC	7.051	1444	5A4
7.241	1483	5CB	7.046	1443	5A3
7.236	1482	5CA	7.041	1442	5A2
7.231	1481	5C9	7.036	1441	5A1
7.227	1480	5C8	7.031	1440	5A0
7.222	1479	5C7	7.026	1439	59F
7.217	1478	5C6	7.021	1438	59E
7.212	1477	5C5	7.017	1437	59D
7.207	1476	5C4	7.012	1436	59C
7.202	1475	5C3	7.007	1435	59B
7.197	1474	5C2	7.002	1434	59A
7.192	1473	5C1	6.997	1433	599
7.187	1472	5C0	6.992	1432	598
7.183	1471	5BF	6.987	1431	597
7.178	1470	5BE	6.982	1430	596
7.173	1469	5BD	6.978	1429	595
7.168	1468	5BC	6.973	1428	594
7.163	1467	5BB	6.968	1427	593
7.158	1466	5BA	6.963	1426	592
7.153	1465	5B9	6.958	1425	591
7.148	1464	5B8	6.953	1424	590
7.144	1463	5B7	6.948	1423	58F
7.139	1462	5B6	6.943	1422	58E
7.134	1461	5B5	6.938	1421	58D
7.129	1460	5B4	6.934	1420	58C
7.124	1459	5B3	6.929	1419	58B
7.119	1458	5B2	6.924	1418	58A
7.114	1457	5B1	6.919	1417	589
7.109	1456	5B0	6.914	1416	588
7.104	1455	5AF	6.909	1415	587
7.100	1454	5AE	6.904	1414	586
7.095	1453	5AD	6.899	1413	585
7.090	1452	5AC	6.895	1412	584
7.085	1451	5AB	6.890	1411	583

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
6.494	1330	532	6.299	1290	50A
6.489	1329	531	6.294	1289	509
6.484	1328	530	6.289	1288	508
6.479	1327	52F	6.284	1287	507
6.475	1326	52E	6.279	1286	506
6.470	1325	52D	6.274	1285	505
6.465	1324	52C	6.270	1284	504
6.460	1323	52B	6.265	1283	503
6.455	1322	52A	6.260	1282	502
6.450	1321	529	6.255	1281	501
6.445	1320	528	6.250	1280	500
6.440	1319	527	6.245	1279	4FF
6.436	1318	526	6.240	1278	4FE
6.431	1317	525	6.235	1277	4FD
6.426	1316	524	6.230	1276	4FC
6.421	1315	523	6.226	1275	4FB
6.416	1314	522	6.221	1274	4FA
6.411	1313	521	6.216	1273	4F9
6.406	1312	520	6.211	1272	4F8
6.401	1311	51F	6.206	1271	4F7
6.396	1310	51E	6.201	1270	4F6
6.392	1309	51D	6.196	1269	4F5
6.387	1308	51C	6.191	1268	4F4
6.382	1307	51B	6.187	1267	4F3
6.377	1306	51A	6.182	1266	4F2
6.372	1305	519	6.177	1265	4F1
6.367	1304	518	6.172	1264	4F0
6.362	1303	517	6.167	1263	4EF
6.357	1302	516	6.162	1262	4EE
6.353	1301	515	6.157	1261	4ED
6.348	1300	514	6.152	1260	4EC
6.343	1299	513	6.147	1259	4EB
6.338	1298	512	6.143	1258	4EA
6.333	1297	511	6.138	1257	4E9
6.328	1296	510	6.133	1256	4E8
6.323	1295	50F	6.128	1255	4E7
6.318	1294	50E	6.123	1254	4E6
6.313	1293	50D	6.118	1253	4E5
6.309	1292	50C	6.113	1252	4E4
6.304	1291	50B	6.108	1251	4E3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
4.932	1010	3F2	4.736	970	3CA
4.927	1009	3F1	4.731	969	3C9
4.922	1008	3F0	4.727	968	3C8
4.917	1007	3EF	4.722	967	3C7
4.912	1006	3EE	4.717	966	3C6
4.907	1005	3ED	4.712	965	3C5
4.902	1004	3EC	4.707	964	3C4
4.897	1003	3EB	4.702	963	3C3
4.893	1002	3EA	4.697	962	3C2
4.888	1001	3E9	4.692	961	3C1
4.883	1000	3E8	4.687	960	3C0
4.878	999	3E7	4.683	959	3BF
4.873	998	3E6	4.678	958	3BE
4.868	997	3E5	4.673	957	3BD
4.863	996	3E4	4.668	956	3BC
4.856	995	3E3	4.663	955	3BB
4.854	994	3E2	4.658	954	3BA
4.849	993	3E1	4.653	953	3B9
4.844	992	3E0	4.648	952	3B8
4.839	991	3DF	4.644	951	3B7
4.834	990	3DE	4.639	950	3B6
4.829	989	3DD	4.634	949	3B5
4.824	988	3DC	4.629	948	3B4
4.819	987	3DB	4.624	947	3B3
4.814	986	3DA	4.619	946	3B2
4.810	985	3D9	4.614	945	3B1
4.805	984	3D8	4.609	944	3B0
4.800	983	3D7	4.604	943	3AF
4.795	982	3D6	4.600	942	3AE
4.790	981	3D5	4.595	941	3AD
4.785	980	3D4	4.590	940	3AC
4.780	979	3D3	4.585	939	3AB
4.775	978	3D2	4.580	938	3AA
4.771	977	3D1	4.575	937	3A9
4.766	976	3D0	4.570	936	3A8
4.761	975	3CF	4.565	935	3A7
4.756	974	3CE	4.561	934	3A6
4.751	973	3CD	4.556	933	3A5
4.746	972	3CC	4.551	932	3A4
4.741	971	3CB	4.546	931	3A3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
4.541	930	3A2	4.346	890	37A
4.536	929	3A1	4.341	889	379
4.531	928	3A0	4.336	888	378
4.526	927	39F	4.331	887	377
4.521	926	39E	4.326	886	376
4.517	925	39D	4.321	885	375
4.512	924	39C	4.316	884	374
4.507	923	39B	4.312	883	373
4.502	922	39A	4.307	882	372
4.497	921	399	4.302	881	371
4.492	920	398	4.297	880	370
4.487	919	397	4.292	879	36F
4.482	918	396	4.287	878	36E
4.478	917	395	4.282	877	36D
4.473	916	394	4.277	876	36C
4.468	915	393	4.272	875	36B
4.463	914	392	4.268	874	36A
4.458	913	391	4.263	873	369
4.453	912	390	4.258	872	368
4.448	911	38F	4.253	871	367
4.443	910	38E	4.248	870	366
4.438	909	38D	4.243	869	365
4.434	908	38C	4.238	868	364
4.429	907	38B	4.233	867	363
4.424	906	38A	4.229	866	362
4.419	905	389	4.224	865	361
4.414	904	388	4.219	864	360
4.409	903	387	4.214	863	35F
4.404	902	386	4.209	862	35E
4.399	901	385	4.204	861	35D
4.395	900	384	4.199	860	35C
4.390	899	383	4.194	859	35B
4.385	898	382	4.189	858	35A
4.380	897	381	4.185	857	359
4.375	896	380	4.180	856	358
4.370	895	37F	4.175	855	357
4.365	894	37E	4.170	854	356
4.360	893	37D	4.165	853	355
4.355	892	37C	4.160	852	354
4.351	891	37B	4.155	851	353

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
2.979	610	262
2.974	609	261
2.969	608	260
2.964	607	25F
2.959	606	25E
2.954	605	25D
2.949	604	25C
2.944	603	25B
2.939	602	25A
2.935	601	259
2.930	600	258
2.925	599	257
2.920	598	256
2.915	597	255
2.910	596	254
2.905	595	253
2.900	594	252
2.896	593	251
2.891	592	250
2.886	591	24F
2.881	590	24E
2.876	589	24D
2.871	588	24C
2.866	587	24B
2.861	586	24A
2.856	585	249
2.852	584	248
2.847	583	247
2.842	582	246
2.837	581	245
2.832	580	244
2.827	579	243
2.822	578	242
2.817	577	241
2.812	576	240
2.808	575	23F
2.803	574	23E
2.798	573	23D
2.793	572	23C
2.788	571	23B

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
2.783	570	23A
2.778	569	239
2.773	568	238
2.769	567	237
2.764	566	236
2.759	565	235
2.754	564	234
2.749	563	233
2.744	562	232
2.739	561	231
2.734	560	230
2.729	559	22F
2.725	558	22E
2.720	557	22D
2.715	556	22C
2.710	555	22B
2.705	554	22A
2.700	553	229
2.695	552	228
2.690	551	227
2.686	550	226
2.681	549	225
2.676	548	224
2.671	547	223
2.666	546	222
2.661	545	221
2.656	544	220
2.651	543	21F
2.646	542	21E
2.642	541	21D
2.637	540	21C
2.632	539	21B
2.627	538	21A
2.622	537	219
2.617	536	218
2.612	535	217
2.607	534	216
2.603	533	215
2.598	532	214
2.593	531	213

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
2.393	490	1EA	2.588	530	212
2.388	489	1E9	2.583	529	211
2.383	488	1E8	2.578	528	210
2.378	487	1E7	2.573	527	20F
2.373	486	1E6	2.568	526	20E
2.368	485	1E5	2.563	525	20D
2.363	484	1E4	2.559	524	20C
2.358	483	1E3	2.554	523	20B
2.354	482	1E2	2.549	522	20A
2.349	481	1E1	2.544	521	209
2.344	480	1E0	2.539	520	208
2.339	479	1DF	2.534	519	207
2.334	478	1DE	2.529	518	206
2.329	477	1DD	2.524	517	205
2.324	476	1DC	2.520	516	204
2.319	475	1DB	2.515	515	203
2.314	474	1DA	2.510	514	202
2.310	473	1D9	2.505	513	201
2.305	472	1D8	2.500	512	200
2.300	471	1D7	2.495	511	1FF
2.295	470	1D6	2.490	510	1FE
2.290	469	1D5	2.485	509	1FD
2.285	468	1D4	2.480	508	1FC
2.280	467	1D3	2.476	507	1FB
2.275	466	1D2	2.471	506	1FA
2.271	465	1D1	2.466	505	1F9
2.266	464	1D0	2.461	504	1F8
2.261	463	1CF	2.456	503	1F7
2.256	462	1CE	2.451	502	1F6
2.251	461	1CD	2.446	501	1F5
2.246	460	1CC	2.441	500	1F4
2.241	459	1CB	2.437	499	1F3
2.236	458	1CA	2.432	498	1F2
2.231	457	1C9	2.427	497	1F1
2.227	456	1C8	2.422	496	1F0
2.222	455	1C7	2.417	495	1EF
2.217	454	1C6	2.412	494	1EE
2.212	453	1C5	2.407	493	1ED
2.207	452	1C4	2.402	492	1EC
2.202	451	1C3	2.397	491	1EB

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
1.807	370	172	1.611	330	14A
1.802	369	171	1.606	329	149
1.797	368	170	1.602	328	148
1.792	367	16F	1.597	327	147
1.787	366	16E	1.592	326	146
1.782	365	16D	1.587	325	145
1.777	364	16C	1.582	324	144
1.772	363	16B	1.577	323	143
1.768	362	16A	1.572	322	142
1.763	361	169	1.567	321	141
1.758	360	168	1.562	320	140
1.753	359	167	1.558	319	13F
1.748	358	166	1.553	318	13E
1.743	357	165	1.548	317	13D
1.738	356	164	1.543	316	13C
1.733	355	163	1.538	315	13B
1.729	354	162	1.533	314	13A
1.724	353	161	1.528	313	139
1.719	352	160	1.523	312	138
1.714	351	15F	1.519	311	137
1.709	350	15E	1.514	310	136
1.704	349	15D	1.509	309	135
1.699	348	15C	1.504	308	134
1.694	347	15B	1.499	307	133
1.689	346	15A	1.494	306	132
1.685	345	159	1.489	305	131
1.680	344	158	1.484	304	130
1.675	343	157	1.479	303	12F
1.670	342	156	1.475	302	12E
1.665	341	155	1.470	301	12D
1.660	340	154	1.465	300	12C
1.655	339	153	1.460	299	12B
1.650	338	152	1.455	298	12A
1.646	337	151	1.450	297	129
1.641	336	150	1.445	296	128
1.636	335	14F	1.440	295	127
1.631	334	14E	1.436	294	126
1.626	333	14D	1.431	293	125
1.621	332	14C	1.426	292	124
1.616	331	14B	1.421	291	123

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
1.025	210	0D2
1.021	209	0D1
1.016	208	0D0
1.011	207	0CF
1.006	206	0CE
1.001	205	0CD
0.996	204	0CC
0.991	203	0CB
0.986	202	0CA
0.981	201	0C9
0.977	200	0C8
0.972	199	0C7
0.967	198	0C6
0.962	197	0C5
0.957	196	0C4
0.952	195	0C3
0.947	194	0C2
0.942	193	0C1
0.937	192	0C0
0.933	191	0BF
0.928	190	0BE
0.923	189	0BD
0.918	188	0BC
0.913	187	0BB
0.908	186	0BA
0.903	185	0B9
0.898	184	0B8
0.894	183	0B7
0.889	182	0B6
0.884	181	0B5
0.879	180	0B4
0.874	179	0B3
0.869	178	0B2
0.864	177	0B1
0.859	176	0B0
0.854	175	0AF
0.850	174	0AE
0.845	173	0AD
0.840	172	0AC
0.835	171	0AB
0.830	170	0AA
0.825	169	0A9
0.820	168	0A8
0.815	167	0A7
0.811	166	0A6
0.806	165	0A5
0.801	164	0A4
0.796	163	0A3
0.791	162	0A2
0.786	161	0A1
0.781	160	0A0
0.776	159	09F
0.771	158	09E
0.767	157	09D
0.762	156	09C
0.757	155	09B
0.752	154	09A
0.747	153	099
0.742	152	098
0.737	151	097
0.732	150	096
0.728	149	095
0.723	148	094
0.718	147	093
0.713	146	092
0.708	145	091
0.703	144	090
0.698	143	08F
0.693	142	08E
0.688	141	08D
0.684	140	08C
0.679	139	08B
0.674	138	08A
0.669	137	089
0.664	136	088
0.659	135	087
0.654	134	086
0.649	133	085
0.645	132	084
0.640	131	083

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
0.635	130	082	0.439	90	05A
0.630	129	081	0.435	89	059
0.625	128	080	0.430	88	058
0.620	127	07F	0.425	87	057
0.615	126	07E	0.420	86	056
0.610	125	07D	0.415	85	055
0.605	124	07C	0.410	84	054
0.601	123	07B	0.405	83	053
0.596	122	07A	0.400	82	052
0.591	121	079	0.396	81	051
0.586	120	078	0.391	80	050
0.581	119	077	0.386	79	04F
0.576	118	076	0.381	78	04E
0.571	117	075	0.376	77	04D
0.566	116	074	0.371	76	04C
0.562	115	073	0.366	75	04B
0.557	114	072	0.361	74	04A
0.552	113	071	0.356	73	049
0.547	112	070	0.352	72	048
0.542	111	06F	0.347	71	047
0.537	110	06E	0.342	70	046
0.532	109	06D	0.337	69	045
0.527	108	06C	0.332	68	044
0.522	107	06B	0.327	67	043
0.518	106	06A	0.322	66	042
0.513	105	069	0.317	65	041
0.508	104	068	0.312	64	040
0.503	103	067	0.308	63	03F
0.498	102	066	0.303	62	03E
0.493	101	065	0.298	61	03D
0.488	100	064	0.293	60	03C
0.483	99	063	0.288	59	03B
0.479	98	062	0.283	58	03A
0.474	97	061	0.278	57	039
0.469	96	060	0.273	56	038
0.464	95	05F	0.269	55	037
0.459	94	05E	0.264	54	036
0.454	93	05D	0.259	53	035
0.449	92	05C	0.254	52	034
0.444	91	05B	0.249	51	033

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-0.146	-30	FE2
-0.151	-31	FE1
-0.156	-32	FE0
-0.161	-33	FDF
-0.166	-34	FDE
-0.171	-35	FDD
-0.176	-36	FDC
-0.181	-37	FDB
-0.186	-38	FDA
-0.190	-39	FD9
-0.195	-40	FD8
-0.200	-41	FD7
-0.205	-42	FD6
-0.210	-43	FD5
-0.215	-44	FD4
-0.220	-45	FD3
-0.225	-46	FD2
-0.229	-47	FD1
-0.234	-48	FD0
-0.239	-49	FCF
-0.244	-50	FCE
-0.249	-51	FCD
-0.254	-52	FCC
-0.259	-53	FCB
-0.264	-54	FCA
-0.269	-55	FC9
-0.273	-56	FC8
-0.278	-57	FC7
-0.283	-58	FC6
-0.288	-59	FC5
-0.293	-60	FC4
-0.298	-61	FC3
-0.303	-62	FC2
-0.308	-63	FC1
-0.312	-64	FC0
-0.317	-65	FBF
-0.322	-66	FBE
-0.327	-67	FBD
-0.332	-68	FBC
-0.337	-69	FBB

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-0.342	-70	FBA
-0.347	-71	FB9
-0.352	-72	FB8
-0.356	-73	FB7
-0.361	-74	FB6
-0.366	-75	FB5
-0.371	-76	FB4
-0.376	-77	FB3
-0.381	-78	FB2
-0.386	-79	FB1
-0.391	-80	FB0
-0.396	-81	FAF
-0.400	-82	FAE
-0.405	-83	FAD
-0.410	-84	FAC
-0.415	-85	FAB
-0.420	-86	FAA
-0.425	-87	FA9
-0.430	-88	FA8
-0.435	-89	FA7
-0.439	-90	FA6
-0.444	-91	FA5
-0.449	-92	FA4
-0.454	-93	FA3
-0.459	-94	FA2
-0.464	-95	FA1
-0.469	-96	FA0
-0.474	-97	F9F
-0.479	-98	F9E
-0.483	-99	F9D
-0.488	-100	F9C
-0.493	-101	F9B
-0.498	-102	F9A
-0.503	-103	F99
-0.508	-104	F98
-0.513	-105	F97
-0.518	-106	F96
-0.522	-107	F95
-0.527	-108	F94
-0.532	-109	F93

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-0.928	-190	F42
-0.933	-191	F41
-0.937	-192	F40
-0.942	-193	F3F
-0.947	-194	F3E
-0.952	-195	F3D
-0.957	-196	F3C
-0.962	-197	F3B
-0.967	-198	F3A
-0.972	-199	F39
-0.977	-200	F38
-0.981	-201	F37
-0.986	-202	F36
-0.991	-203	F35
-0.996	-204	F34
-1.001	-205	F33
-1.006	-206	F32
-1.011	-207	F31
-1.016	-208	F30
-1.021	-209	F2F
-1.025	-210	F2E
-1.030	-211	F2D
-1.035	-212	F2C
-1.040	-213	F2B
-1.045	-214	F2A
-1.050	-215	F29
-1.055	-216	F28
-1.060	-217	F27
-1.064	-218	F26
-1.069	-219	F25
-1.074	-220	F24
-1.079	-221	F23
-1.084	-222	F22
-1.089	-223	F21
-1.094	-224	F20
-1.099	-225	F1F
-1.104	-226	F1E
-1.108	-227	F1D
-1.113	-228	F1C
-1.118	-229	F1B

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-1.123	-230	F1A
-1.128	-231	F19
-1.133	-232	F18
-1.138	-233	F17
-1.143	-234	F16
-1.147	-235	F15
-1.152	-236	F14
-1.157	-237	F13
-1.162	-238	F12
-1.167	-239	F11
-1.172	-240	F10
-1.177	-241	F0F
-1.182	-242	F0E
-1.187	-243	F0D
-1.191	-244	F0C
-1.196	-245	F0B
-1.201	-246	F0A
-1.206	-247	F09
-1.211	-248	F08
-1.216	-249	F07
-1.221	-250	F06
-1.226	-251	F05
-1.230	-252	F04
-1.235	-253	F03
-1.240	-254	F02
-1.245	-255	F01
-1.250	-256	F00
-1.255	-257	EFF
-1.260	-258	EFE
-1.265	-259	EFD
-1.270	-260	EFC
-1.274	-261	EFB
-1.279	-262	EFA
-1.284	-263	EF9
-1.289	-264	EF8
-1.294	-265	EF7
-1.299	-266	EF6
-1.304	-267	EF5
-1.309	-268	EF4
-1.313	-269	EF3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-2.490	-510	E02
-2.495	-511	E01
-2.500	-512	E00
-2.505	-513	DFF
-2.510	-514	DFE
-2.515	-515	DFD
-2.520	-516	DFC
-2.524	-517	DFB
-2.529	-518	DFA
-2.534	-519	DF9
-2.539	-520	DF8
-2.544	-521	DF7
-2.549	-522	DF6
-2.554	-523	DF5
-2.559	-524	DF4
-2.563	-525	DF3
-2.568	-526	DF2
-2.573	-527	DF1
-2.578	-528	DF0
-2.583	-529	DEF
-2.588	-530	DEE
-2.593	-531	DED
-2.598	-532	DEC
-2.603	-533	DEB
-2.607	-534	DEA
-2.612	-535	DE9
-2.617	-536	DE8
-2.622	-537	DE7
-2.627	-538	DE6
-2.632	-539	DE5
-2.637	-540	DE4
-2.642	-541	DE3
-2.646	-542	DE2
-2.651	-543	DE1
-2.656	-544	DE0
-2.661	-545	DDF
-2.666	-546	DDE
-2.671	-547	DDD
-2.676	-548	DDC
-2.681	-549	DDB

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-2.686	-550	DDA
-2.690	-551	DD9
-2.695	-552	DD8
-2.700	-553	DD7
-2.705	-554	DD6
-2.710	-555	DD5
-2.715	-556	DD4
-2.720	-557	DD3
-2.725	-558	DD2
-2.729	-559	DD1
-2.734	-560	DD0
-2.739	-561	DCF
-2.744	-562	DCE
-2.749	-563	DCD
-2.754	-564	DCC
-2.759	-565	DCB
-2.764	-566	DCA
-2.769	-567	DC9
-2.773	-568	DC8
-2.778	-569	DC7
-2.783	-570	DC6
-2.788	-571	DC5
-2.793	-572	DC4
-2.798	-573	DC3
-2.803	-574	DC2
-2.808	-575	DC1
-2.812	-576	DC0
-2.817	-577	DBF
-2.822	-578	DBE
-2.827	-579	DBD
-2.832	-580	DBC
-2.837	-581	DBB
-2.842	-582	DBA
-2.847	-583	DB9
-2.852	-584	DB8
-2.856	-585	DB7
-2.861	-586	DB6
-2.866	-587	DB5
-2.871	-588	DB4
-2.876	-589	DB3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-3.271	-670	D62
-3.276	-671	D61
-3.281	-672	D60
-3.286	-673	D5F
-3.291	-674	D5E
-3.296	-675	D5D
-3.301	-676	D5C
-3.306	-677	D5B
-3.311	-678	D5A
-3.315	-679	D59
-3.320	-680	D58
-3.325	-681	D57
-3.330	-682	D56
-3.335	-683	D55
-3.340	-684	D54
-3.345	-685	D53
-3.350	-686	D52
-3.354	-687	D51
-3.359	-688	D50
-3.364	-689	D4F
-3.369	-690	D4E
-3.374	-691	D4D
-3.379	-692	D4C
-3.384	-693	D4B
-3.389	-694	D4A
-3.394	-695	D49
-3.398	-696	D48
-3.403	-697	D47
-3.408	-698	D46
-3.413	-699	D45
-3.418	-700	D44
-3.423	-701	D43
-3.428	-702	D42
-3.433	-703	D41
-3.437	-704	D40
-3.442	-705	D3F
-3.447	-706	D3E
-3.452	-707	D3D
-3.457	-708	D3C
-3.462	-709	D3B

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-3.467	-710	D3A
-3.472	-711	D39
-3.477	-712	D38
-3.481	-713	D37
-3.486	-714	D36
-3.491	-715	D35
-3.496	-716	D34
-3.501	-717	D33
-3.506	-718	D32
-3.511	-719	D31
-3.516	-720	D30
-3.521	-721	D2F
-3.525	-722	D2E
-3.530	-723	D2D
-3.535	-724	D2C
-3.540	-725	D2B
-3.545	-726	D2A
-3.550	-727	D29
-3.555	-728	D28
-3.560	-729	D27
-3.564	-730	D26
-3.569	-731	D25
-3.574	-732	D24
-3.579	-733	D23
-3.584	-734	D22
-3.589	-735	D21
-3.594	-736	D20
-3.599	-737	D1F
-3.604	-738	D1E
-3.608	-739	D1D
-3.613	-740	D1C
-3.618	-741	D1B
-3.623	-742	D1A
-3.628	-743	D19
-3.633	-744	D18
-3.638	-745	D17
-3.643	-746	D16
-3.647	-747	D15
-3.652	-748	D14
-3.657	-749	D13

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-4.834	-990	C22
-4.839	-991	C21
-4.844	-992	C20
-4.849	-993	C1F
-4.854	-994	C1E
-4.858	-995	C1D
-4.863	-996	C1C
-4.868	-997	C1B
-4.873	-998	C1A
-4.878	-999	C19
-4.883	-1000	C18
-4.888	-1001	C17
-4.893	-1002	C16
-4.897	-1003	C15
-4.902	-1004	C14
-4.907	-1005	C13
-4.912	-1006	C12
-4.917	-1007	C11
-4.922	-1008	C10
-4.927	-1009	C0F
-4.932	-1010	C0E
-4.937	-1011	C0D
-4.941	-1012	C0C
-4.946	-1013	C0B
-4.951	-1014	C0A
-4.956	-1015	C09
-4.961	-1016	C08
-4.966	-1017	C07
-4.971	-1018	C06
-4.976	-1019	C05
-4.980	-1020	C04
-4.985	-1021	C03
-4.990	-1022	C02
-4.995	-1023	C01
-5.000	-1024	C00
-5.005	-1025	BFF
-5.010	-1026	BFE
-5.015	-1027	BFD
-5.020	-1028	BFC
-5.024	-1029	BFB

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-5.029	-1030	BFA
-5.034	-1031	BF9
-5.039	-1032	BF8
-5.044	-1033	BF7
-5.049	-1034	BF6
-5.054	-1035	BF5
-5.059	-1036	BF4
-5.063	-1037	BF3
-5.068	-1038	BF2
-5.073	-1039	BF1
-5.078	-1040	BF0
-5.083	-1041	BEF
-5.088	-1042	BEE
-5.093	-1043	BED
-5.098	-1044	BEC
-5.103	-1045	BEB
-5.107	-1046	BEA
-5.112	-1047	BE9
-5.117	-1048	BE8
-5.122	-1049	BE7
-5.127	-1050	BE6
-5.132	-1051	BE5
-5.137	-1052	BE4
-5.142	-1053	BE3
-5.146	-1054	BE2
-5.151	-1055	BE1
-5.156	-1056	BE0
-5.161	-1057	BDF
-5.166	-1058	BDE
-5.171	-1059	BDD
-5.176	-1060	BDC
-5.181	-1061	BDB
-5.186	-1062	BDA
-5.190	-1063	BD9
-5.195	-1064	BD8
-5.200	-1065	BD7
-5.205	-1066	BD6
-5.210	-1067	BD5
-5.215	-1068	BD4
-5.220	-1069	BD3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-5.225	-1070	8D2
-5.229	-1071	8D1
-5.234	-1072	8D0
-5.239	-1073	8CF
-5.244	-1074	8CE
-5.249	-1075	8CD
-5.254	-1076	8CC
-5.259	-1077	8CB
-5.264	-1078	8CA
-5.269	-1079	8C9
-5.273	-1080	8C8
-5.278	-1081	8C7
-5.283	-1082	8C6
-5.288	-1083	8C5
-5.293	-1084	8C4
-5.298	-1085	8C3
-5.303	-1086	8C2
-5.308	-1087	8C1
-5.312	-1088	8C0
-5.317	-1089	8BF
-5.322	-1090	8BE
-5.327	-1091	8BD
-5.332	-1092	8BC
-5.337	-1093	8BB
-5.342	-1094	8BA
-5.347	-1095	8B9
-5.352	-1096	8B8
-5.356	-1097	8B7
-5.361	-1098	8B6
-5.366	-1099	8B5
-5.371	-1100	8B4
-5.376	-1101	8B3
-5.381	-1102	8B2
-5.386	-1103	8B1
-5.391	-1104	8B0
-5.396	-1105	8AF
-5.400	-1106	8AE
-5.405	-1107	8AD
-5.410	-1108	8AC
-5.415	-1109	8AB

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-5.420	-1110	8AA
-5.425	-1111	8A9
-5.430	-1112	8A8
-5.435	-1113	8A7
-5.439	-1114	8A6
-5.444	-1115	8A5
-5.449	-1116	8A4
-5.454	-1117	8A3
-5.459	-1118	8A2
-5.464	-1119	8A1
-5.469	-1120	8A0
-5.474	-1121	89F
-5.479	-1122	89E
-5.483	-1123	89D
-5.488	-1124	89C
-5.493	-1125	89B
-5.498	-1126	89A
-5.503	-1127	899
-5.508	-1128	898
-5.513	-1129	897
-5.518	-1130	896
-5.522	-1131	895
-5.527	-1132	894
-5.532	-1133	893
-5.537	-1134	892
-5.542	-1135	891
-5.547	-1136	890
-5.552	-1137	88F
-5.557	-1138	88E
-5.562	-1139	88D
-5.566	-1140	88C
-5.571	-1141	88B
-5.576	-1142	88A
-5.581	-1143	889
-5.586	-1144	888
-5.591	-1145	887
-5.596	-1146	886
-5.601	-1147	885
-5.605	-1148	884
-5.610	-1149	883

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-5.615	-1150	882
-5.620	-1151	881
-5.625	-1152	880
-5.630	-1153	87F
-5.635	-1154	87E
-5.640	-1155	87D
-5.645	-1156	87C
-5.649	-1157	87B
-5.654	-1158	87A
-5.659	-1159	879
-5.664	-1160	878
-5.669	-1161	877
-5.674	-1162	876
-5.679	-1163	875
-5.684	-1164	874
-5.688	-1165	873
-5.693	-1166	872
-5.698	-1167	871
-5.703	-1168	870
-5.708	-1169	86F
-5.713	-1170	86E
-5.718	-1171	86D
-5.723	-1172	86C
-5.728	-1173	86B
-5.732	-1174	86A
-5.737	-1175	869
-5.742	-1176	868
-5.747	-1177	867
-5.752	-1178	866
-5.757	-1179	865
-5.762	-1180	864
-5.767	-1181	863
-5.771	-1182	862
-5.776	-1183	861
-5.781	-1184	860
-5.786	-1185	85F
-5.791	-1186	85E
-5.796	-1187	85D
-5.801	-1188	85C
-5.806	-1189	85B

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-5.811	-1190	85A
-5.815	-1191	859
-5.820	-1192	858
-5.825	-1193	857
-5.830	-1194	856
-5.835	-1195	855
-5.840	-1196	854
-5.845	-1197	853
-5.850	-1198	852
-5.854	-1199	851
-5.859	-1200	850
-5.864	-1201	84F
-5.869	-1202	84E
-5.874	-1203	84D
-5.879	-1204	84C
-5.884	-1205	84B
-5.889	-1206	84A
-5.894	-1207	849
-5.898	-1208	848
-5.903	-1209	847
-5.908	-1210	846
-5.913	-1211	845
-5.918	-1212	844
-5.923	-1213	843
-5.928	-1214	842
-5.933	-1215	841
-5.937	-1216	840
-5.942	-1217	83F
-5.947	-1218	83E
-5.952	-1219	83D
-5.957	-1220	83C
-5.962	-1221	83B
-5.967	-1222	83A
-5.972	-1223	839
-5.977	-1224	838
-5.981	-1225	837
-5.986	-1226	836
-5.991	-1227	835
-5.996	-1228	834
-6.001	-1229	833

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-6.396	-1310	AE2	-6.592	-1350	ABA
-6.401	-1311	AE1	-6.597	-1351	AB9
-6.406	-1312	AE0	-6.602	-1352	AB8
-6.411	-1313	ADF	-6.606	-1353	AB7
-6.416	-1314	ADE	-6.611	-1354	AB6
-6.421	-1315	ADD	-6.616	-1355	AB5
-6.426	-1316	ADC	-6.621	-1356	AB4
-6.431	-1317	ADB	-6.626	-1357	AB3
-6.436	-1318	ADA	-6.631	-1358	AB2
-6.440	-1319	AD9	-6.636	-1359	AB1
-6.445	-1320	AD8	-6.641	-1360	AB0
-6.450	-1321	AD7	-6.646	-1361	AAF
-6.455	-1322	AD6	-6.650	-1362	AAE
-6.460	-1323	AD5	-6.655	-1363	AAD
-6.465	-1324	AD4	-6.660	-1364	AAC
-6.470	-1325	AD3	-6.665	-1365	AAB
-6.475	-1326	AD2	-6.670	-1366	AAA
-6.479	-1327	AD1	-6.675	-1367	AA9
-6.484	-1328	AD0	-6.680	-1368	AA8
-6.489	-1329	ACF	-6.685	-1369	AA7
-6.494	-1330	ACE	-6.689	-1370	AA6
-6.499	-1331	ACD	-6.694	-1371	AA5
-6.504	-1332	ACC	-6.699	-1372	AA4
-6.509	-1333	ACB	-6.704	-1373	AA3
-6.514	-1334	ACA	-6.709	-1374	AA2
-6.519	-1335	AC9	-6.714	-1375	AA1
-6.523	-1336	AC8	-6.719	-1376	AA0
-6.528	-1337	AC7	-6.724	-1377	A9F
-6.533	-1338	AC6	-6.729	-1378	A9E
-6.538	-1339	AC5	-6.733	-1379	A9D
-6.543	-1340	AC4	-6.738	-1380	A9C
-6.548	-1341	AC3	-6.743	-1381	A9B
-6.553	-1342	AC2	-6.748	-1382	A9A
-6.558	-1343	AC1	-6.753	-1383	A99
-6.562	-1344	AC0	-6.758	-1384	A98
-6.567	-1345	ABF	-6.763	-1385	A97
-6.572	-1346	ABE	-6.768	-1386	A96
-6.577	-1347	ABD	-6.772	-1387	A95
-6.582	-1348	ABC	-6.777	-1388	A94
-6.587	-1349	ABB	-6.782	-1389	A93

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-7.178	-1470	A42
-7.183	-1471	A41
-7.187	-1472	A40
-7.192	-1473	A3F
-7.197	-1474	A3E
-7.202	-1475	A3D
-7.207	-1476	A3C
-7.212	-1477	A3B
-7.217	-1478	A3A
-7.222	-1479	A39
-7.227	-1480	A38
-7.231	-1481	A37
-7.236	-1482	A36
-7.241	-1483	A35
-7.246	-1484	A34
-7.251	-1485	A33
-7.256	-1486	A32
-7.261	-1487	A31
-7.266	-1488	A30
-7.271	-1489	A2F
-7.275	-1490	A2E
-7.280	-1491	A2D
-7.285	-1492	A2C
-7.290	-1493	A2B
-7.295	-1494	A2A
-7.300	-1495	A29
-7.305	-1496	A28
-7.310	-1497	A27
-7.314	-1498	A26
-7.319	-1499	A25
-7.324	-1500	A24
-7.329	-1501	A23
-7.334	-1502	A22
-7.339	-1503	A21
-7.344	-1504	A20
-7.349	-1505	A1F
-7.354	-1506	A1E
-7.358	-1507	A1D
-7.363	-1508	A1C
-7.368	-1509	A1B

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-7.373	-1510	A1A
-7.378	-1511	A19
-7.383	-1512	A18
-7.388	-1513	A17
-7.393	-1514	A16
-7.397	-1515	A15
-7.402	-1516	A14
-7.407	-1517	A13
-7.412	-1518	A12
-7.417	-1519	A11
-7.422	-1520	A10
-7.427	-1521	A0F
-7.432	-1522	A0E
-7.437	-1523	A0D
-7.441	-1524	A0C
-7.446	-1525	A0B
-7.451	-1526	A0A
-7.456	-1527	A09
-7.461	-1528	A08
-7.466	-1529	A07
-7.471	-1530	A06
-7.476	-1531	A05
-7.480	-1532	A04
-7.485	-1533	A03
-7.490	-1534	A02
-7.495	-1535	A01
-7.500	-1536	A00
-7.505	-1537	9FF
-7.510	-1538	9FE
-7.515	-1539	9FD
-7.520	-1540	9FC
-7.524	-1541	9FB
-7.529	-1542	9FA
-7.534	-1543	9F9
-7.539	-1544	9F8
-7.544	-1545	9F7
-7.549	-1546	9F6
-7.554	-1547	9F5
-7.559	-1548	9F4
-7.563	-1549	9F3

DC VOLTS	DECIMAL COUNTS	HEX COUNTS	DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-8.350	-1710	952	-8.545	-1750	92A
-8.354	-1711	951	-8.550	-1751	929
-8.359	-1712	950	-8.555	-1752	928
-8.364	-1713	94F	-8.560	-1753	927
-8.369	-1714	94E	-8.564	-1754	926
-8.374	-1715	94D	-8.569	-1755	925
-8.379	-1716	94C	-8.574	-1756	924
-8.384	-1717	94B	-8.579	-1757	923
-8.389	-1718	94A	-8.584	-1758	922
-8.394	-1719	949	-8.589	-1759	921
-8.398	-1720	948	-8.594	-1760	920
-8.403	-1721	947	-8.599	-1761	91F
-8.408	-1722	946	-8.604	-1762	91E
-8.413	-1723	945	-8.608	-1763	91D
-8.418	-1724	944	-8.613	-1764	91C
-8.423	-1725	943	-8.618	-1765	91B
-8.428	-1726	942	-8.623	-1766	91A
-8.433	-1727	941	-8.628	-1767	919
-8.437	-1728	940	-8.633	-1768	918
-8.442	-1729	93F	-8.638	-1769	917
-8.447	-1730	93E	-8.643	-1770	916
-8.452	-1731	93D	-8.647	-1771	915
-8.457	-1732	93C	-8.652	-1772	914
-8.462	-1733	93B	-8.657	-1773	913
-8.467	-1734	93A	-8.662	-1774	912
-8.472	-1735	939	-8.667	-1775	911
-8.477	-1736	938	-8.672	-1776	910
-8.481	-1737	937	-8.677	-1777	90F
-8.486	-1738	936	-8.682	-1778	90E
-8.491	-1739	935	-8.687	-1779	90D
-8.496	-1740	934	-8.691	-1780	90C
-8.501	-1741	933	-8.696	-1781	90B
-8.506	-1742	932	-8.701	-1782	90A
-8.511	-1743	931	-8.706	-1783	909
-8.516	-1744	930	-8.711	-1784	908
-8.521	-1745	92F	-8.716	-1785	907
-8.525	-1746	92E	-8.721	-1786	906
-8.530	-1747	92D	-8.726	-1787	905
-8.535	-1748	92C	-8.730	-1788	904
-8.540	-1749	92B	-8.735	-1789	903

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-9.131	-1870	8B2
-9.136	-1871	8B1
-9.141	-1872	8B0
-9.146	-1873	8AF
-9.150	-1874	8AE
-9.155	-1875	8AD
-9.160	-1876	8AC
-9.165	-1877	8AB
-9.170	-1878	8AA
-9.175	-1879	8A9
-9.180	-1880	8A8
-9.185	-1881	8A7
-9.189	-1882	8A6
-9.194	-1883	8A5
-9.199	-1884	8A4
-9.204	-1885	8A3
-9.209	-1886	8A2
-9.214	-1887	8A1
-9.219	-1888	8A0
-9.224	-1889	89F
-9.229	-1890	89E
-9.233	-1891	89D
-9.238	-1892	89C
-9.243	-1893	89B
-9.248	-1894	89A
-9.253	-1895	899
-9.258	-1896	898
-9.263	-1897	897
-9.268	-1898	896
-9.272	-1899	895
-9.277	-1900	894
-9.282	-1901	893
-9.287	-1902	892
-9.292	-1903	891
-9.297	-1904	890
-9.302	-1905	88F
-9.307	-1906	88E
-9.312	-1907	88D
-9.316	-1908	88C
-9.321	-1909	88B

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-9.326	-1910	88A
-9.331	-1911	889
-9.336	-1912	888
-9.341	-1913	887
-9.346	-1914	886
-9.351	-1915	885
-9.355	-1916	884
-9.360	-1917	883
-9.365	-1918	882
-9.370	-1919	881
-9.375	-1920	880
-9.380	-1921	87F
-9.385	-1922	87E
-9.390	-1923	87D
-9.395	-1924	87C
-9.399	-1925	87B
-9.404	-1926	87A
-9.409	-1927	879
-9.414	-1928	878
-9.419	-1929	877
-9.424	-1930	876
-9.429	-1931	875
-9.434	-1932	874
-9.438	-1933	873
-9.443	-1934	872
-9.448	-1935	871
-9.453	-1936	870
-9.458	-1937	86F
-9.463	-1938	86E
-9.468	-1939	86D
-9.473	-1940	86C
-9.478	-1941	86B
-9.482	-1942	86A
-9.487	-1943	869
-9.492	-1944	868
-9.497	-1945	867
-9.502	-1946	866
-9.507	-1947	865
-9.512	-1948	864
-9.517	-1949	863

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-9.521	-1950	862
-9.526	-1951	861
-9.531	-1952	860
-9.536	-1953	85F
-9.541	-1954	85E
-9.546	-1955	85D
-9.551	-1956	85C
-9.556	-1957	85B
-9.561	-1958	85A
-9.565	-1959	859
-9.570	-1960	858
-9.575	-1961	857
-9.580	-1962	856
-9.585	-1963	855
-9.590	-1964	854
-9.595	-1965	853
-9.600	-1966	852
-9.604	-1967	851
-9.609	-1968	850
-9.614	-1969	84F
-9.619	-1970	84E
-9.624	-1971	84D
-9.629	-1972	84C
-9.634	-1973	84B
-9.639	-1974	84A
-9.644	-1975	849
-9.648	-1976	848
-9.653	-1977	847
-9.658	-1978	846
-9.663	-1979	845
-9.668	-1980	844
-9.673	-1981	843
-9.678	-1982	842
-9.683	-1983	841
-9.687	-1984	840
-9.692	-1985	83F
-9.697	-1986	83E
-9.702	-1987	83D
-9.707	-1988	83C
-9.712	-1989	83B

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-9.717	-1990	83A
-9.722	-1991	839
-9.727	-1992	838
-9.731	-1993	837
-9.736	-1994	836
-9.741	-1995	835
-9.746	-1996	834
-9.751	-1997	833
-9.756	-1998	832
-9.761	-1999	831
-9.766	-2000	830
-9.771	-2001	82F
-9.775	-2002	82E
-9.780	-2003	82D
-9.785	-2004	82C
-9.790	-2005	82B
-9.795	-2006	82A
-9.800	-2007	829
-9.805	-2008	828
-9.810	-2009	827
-9.814	-2010	826
-9.819	-2011	825
-9.824	-2012	824
-9.829	-2013	823
-9.834	-2014	822
-9.839	-2015	821
-9.844	-2016	820
-9.849	-2017	81F
-9.854	-2018	81E
-9.858	-2019	81D
-9.863	-2020	81C
-9.868	-2021	81B
-9.873	-2022	81A
-9.878	-2023	819
-9.883	-2024	818
-9.888	-2025	817
-9.893	-2026	816
-9.897	-2027	815
-9.902	-2028	814
-9.907	-2029	813

DC VOLTS	DECIMAL COUNTS	HEX COUNTS
-9.912	-2030	812
-9.917	-2031	811
-9.922	-2032	810
-9.927	-2033	80F
-9.932	-2034	80E
-9.937	-2035	80D
-9.941	-2036	80C
-9.946	-2037	80B
-9.951	-2038	80A
-9.956	-2039	809
-9.961	-2040	808
-9.966	-2041	807
-9.971	-2042	806
-9.976	-2043	805
-9.980	-2044	804
-9.985	-2045	803
-9.990	-2046	802
-9.995	-2047	801
-10.000	-2048	800

APPENDIX 3

RESISTANCE TEMPERATURE DETECTOR (RTD) TEMPERATURE TABLE

This table provides decimal and hexadecimal RTD readings for specified values of ambient temperature.

AMBIENT TEMPERATURE:		DECIMAL	HEX
DEGF	DEGC	+/-5 CNTS	+/-5 CNTS
60	15.5	226	0E2
61	16.1	213	005
62	16.6	200	0C8
63	17.2	186	0BA
64	17.7	173	0AD
65	18.3	160	0A0
66	18.8	147	093
67	19.4	133	085
68	20.0	120	078
69	20.5	107	06B
70	21.1	93	05B
71	21.7	80	050
72	22.2	67	043
73	22.8	53	035
74	23.3	40	028
75	23.8	27	018
76	24.4	13	00D
77	25.0	0	000
78	25.5	-13	FF3
79	26.1	-27	FE5
80	26.7	-40	FD8
81	27.2	-53	FCB
82	27.8	-67	FBD
83	28.3	-80	FB0
84	28.8	-93	FA5
85	29.4	-107	F95
86	30.0	-120	F88
87	30.5	-133	F78
88	31.1	-147	F6D
89	31.6	-160	F60
91	32.8	-173	F53
92	33.3	-186	F46
93	33.8	-200	F38
94	34.4	-213	F2B
95	35.0	-226	F1E
96	35.5	-240	F10
97	36.1	-253	F03
98	36.6	-266	EF6
99	37.2	-280	EE8

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